ADVANCED CALCULUS AND PROBABILITY

Course Code: MS 8227

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COLLEGE: Science and Technical Education
DEPARTMENT: Mathematics and Statistics

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Mode of Assessment

Table 1: Mode of Assessment

CA & SE	WEEK	MARKS
Test 01	7^{th}	15%
Assignment	9^{th}	10%
Test 02	10^{th}	15%
Semester Exam	15^{th}	60%

Learning outcome

- Ability to use a range of mathematical and statistical techniques and concepts:
- Ability to use concepts of descriptive statistics and data presentation and interpretation.
- Ability to explain the basic principles of probability.
- Ability to apply knowledge of advanced calculus in solving physical and engineering problems
- Ability to apply knowledge Fourier in solving engineering problems

Course Contents

- Descriptive statistics
- Basic probability
- Discrete Outcomes
- Continuous outcomes
- Fourier series and Transforms
- Line and Multiple Integrals

Introduction

- Statistics deals with collection, analysis and interpretation of numerical data.
  Example of numerical data.
  (i) Marks obtained by students after doing a subject test.
  (ii) Ages of students in a school
  (iii) wages of workers in a company etc
- Statistics can be used to;
  (i) find total size of population.
  (ii) predict the future by analyzing the past events.
  (iii)find relationship between variables. For example, height and weight.

Definition of key terms

1. Raw Data

- Data collected which has not yet organized numerically.
- Data that has not been processed for use.
- Data that comes straight off the experiment.
- Data represented exactly as it was captured at its source without transformation, aggregation or calculation
- Raw data sometimes known as source data or atomic data.

example of raw data

i The marks obtained by 10 students in a test. 6, 7, 5, 9, 1, 8, 8, 2, 7, 3 .

ii Commercial transactions such as a purchase or return

Discrete data and continuous data

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- Discrete data assume only exact values

example discrete data

Figure 1: Number of students in different classes

example of discrete data

Figure 2: Number of cars passing a certain point after every 15min.

Continuous data

- Data which assume intermediate values.

example of continuous data

Figure 3: Weight s of students in a class

example 2

- Ages of pupils in a primary school etc.

Descriptive statistics

Is the method of describing set of data element whether in population or in a sample

Population

is a complete set of data element in statistics. Population does not mean people only, it can be a population of trees, stones etc.

Sample

A portion of a population selected for further analysis

Statistic

A summary typically numerical of a variable over a sample or characteristics of a sample.

Range

- The difference between greatest and smallest observations.
- Consider the marks: 30%, 70%, 50%, and 90%.
- Range = 60%.

Frequency

- The number of occurrence of a particular value.

Example

Consider the numbers: 5, 6 , 7, 8, 6, 5, 10, 7, 6

Number	5	6	7	8	10
Frequency	2	3	2	1	1

Frequency distribution table

Frequency distribution table

- A tabular arrangement which shows data and their corresponding frequencies.

- Relative frequency$= \dfrac{\text{Observed frequency}}{\text{Total frequency}}$

Example

Number	5	6	7	8	10
Frequency	2	3	2	1	1
Rel.Freq	$\frac{2}{9} = 22.22\%$	$\frac{3}{9} = 33.33\%$	$\frac{2}{9} = 22.22\%$	$\frac{1}{9} = 11.11\%$	$\frac{1}{9} = 11.11\%$

The relative frequency is generally expressed in percentage.

Class limits

Let's consider the interval 9 – 11

- The end numbers 9 and 11 are called class limits.
- The smallest number 9 is the lower class limit.
- The great number 11 is the upper class limit.
- The class interval 9 – 11 theoretically includes all marks (measurements) from 8.5 – 11.5.
- The number 8.5 and 11.5 are called boundaries or true class limits.
- 8.5 is the lower class boundary.
- 11.5 is the upper class boundary.
- How to get class boundary.
- Class boundary $= \dfrac{\text{U.c.l of nth class + L.c.l of (n +1)th class}}{2}$

Class size

- The difference between lower and upper class boundaries.

Example

For the class interval 9 – 11 has class size of 3.

Note: If all class intervals of frequency distribution has equal widths then;
class Size $=$ the difference between successive lower class limits or successive upper class limits.

Class mark or class mid-point

- The mid-point of the class-interval.

Marks	30-39	40-49	50-59	60-69	70-79
Frequency	10	14	26	20	18
Class boundary	29.5-39.5	39.5-49.5	49.5-59.5	59.5-69.5	69.5-79.5
Class mark	34.5	44.5	54.5	64.5	74.5

- Upper class boundary$= \dfrac{39+40}{2} = 39.5$

- Class Mark of 1^{st} class interval $= \dfrac{30+39}{2} = 34.5$

Note:
If all class intervals have equal size then
class size = common difference between successive class marks

example

From the table above
Class size $= 44.5 - 34.5 = 10$.
The class boundaries are mid-way between class-marks.

- Refer to table 3. u.c.b of class-mark $34.5 = \dfrac{34.5+44.5}{2} = 39.5$

- l.c.b of 1st class-interval$= 39.5 - 10 = 29.5$

Example

Example

If the class marks in the frequency distribution of lengths of certain leaves are 128, 137, 146, 155, 164, 173 and 182. Find
a) the class size.
b) the class boundaries.
c) the class limits assuming that the length were measured to the nearest millimetre.

Solution

a) Class size$= 137 - 128 = 9$.
b) Class-boundaries
u.c.b of 1^{st} class mark$= \dfrac{128+137}{2} = 132.5$
l.c.b of 1^{st} class mark$= 132.5 - 9 = 123.5$

Solution

(c) Solution

Class Marks	Class boundary	Class limits	Frequency
128	123.5 - 132.5	124 – 132	1
137	132.5 – 141.5	133 – 141	1
146	141.5 – 150.5	142 – 150	1
155	150.5 - 159.5	151 – 159	1
164	159.5 -168.5	160 – 168	1
173	168.5 - 177.5	169 – 177	1
182	177.5 – 186.5	178 – 186	1

- Data summarized in classes, for example 124 – 132, is called grouped data.
  When discrete data is arranged into classes certain original information is lost.

Continuous frequency distribution

- C.f.d is the distribution formed by Continuous variable/data.Example of c.v: age, height, length.
- If the variable is age, we don't take class intervals as 15 – 19, 20 – 24 and so on. Why?
- Because the persons with ages between 19 and 20 are not taken into consideration.

Example

In such a case we form class-intervals as follows:
Age in years
Below 15
15 or more but < 20.
20 or more but < 25.
25 or more but < 30.
And so on.

This way all persons with any fraction of age are included in one group or the other.

Example

For practical purpose we write the above classes as
15 - 20
20 - 25
25 - 30
30 - 35
and so on.

Example

The following are the results in mathematics test of twenty students.
27, 21, 24, 27, 31, 40, 45, 46, 50, 40, 38, 29, 49, 48, 35, 34, 25, 26, 35, 25.
Prepare the frequency distribution by using class-Intervals, 21 - 25, 26 - 30, 31 - 35,.........

example

The following are results of English examination of 50 students in ABC school.
21, 23, 48, 54, 64, 77, 68, 59, 31, 46, 33, 53, 61, 71, 82, 75, 61, 64, 34, 25, 26, 31, 32, 36, 48, 45, 45, 44, 55, 54, 60, 67, 71, 74, 37, 38, 34, 29, 40, 41, 26, 42, 78, 80, 86, 90, 43, 27, 45, 67.
Draw a frequency distribution by using class marks 23, 28, 33, ...

Solution

- Given the class marks 23, 28, 33, ...
- Upper class boundary of class mark 23$= \dfrac{23+28}{2} = 25.5$
- Class-size $= 28 - 23 = 5$
- Lower class boundary of class mark 23 $= 25.5 - 5 = 20.5$

solution

Class Interval	Frequency	Class mark
21-25	3	23
26-30	4	28
31-35	6	33
36-40	4	38
42-45	7	43
46-50	3	48
51-55	4	53
56-60	2	58
61-65	4	63
66-70	3	68
71-75	4	73
76-80	3	78
81-85	1	83
86-90	2	88

Example

The data below gives the time in minutes it takes a commuter to drive to work for a period lasting 50 days.
25, 27, 43, 23, 39, 33, 29, 26, 34, 22, 28, 30, 39, 32, 25, 27, 28, 28, 27, 35, 28, 46, 24, 24, 22, 31, 28, 27, 31, 22, 32, 36, 22, 26, 34, 30, 37, 35, 42, 25, 37, 30, 27, 31, 30, 48, 28, 24, 25.
Construct the freq. distr.table having six classes for which 20 is the lower limit of a first class and 49 is the upper limit of the sixth class.

Example

Make a frequency table using five classes for the following data.
60, 25, 29, 25, 40, 35, 50, 80, 80, 90, 80, 55, 70, 40, 100, 52, 30, 70, 45, 60.

Assignment

Discuss the methods of data collection.

Write short notes on histogram, frequency polygon, cumulative frequency curve(ogive), pie and bar charts. Show how to apply the graphical data representations by solving one question each. Date of submission.

Describe sampling techniques. Date of submission 13/11/2023.

Measures of central tendency

Measures of central tendency

- A measure of central tendency is a value that describes a set of data by identifying the central position of a given set of values.
- The central tendency of a statistical value means a typical value around which all other values tend to concentrate.
- Each in a different sense tells what value of the data is clustered around
- A measure of central tendency indicates the location or the general position of the distribution on the x-axis.
- It is also known as a measure of location or position.

Measures of central tendency cont...

These includes;

- Mean
- Mode
- Median

Arithmetic Mean

- Arithmetic mean $\bar{x}$ is the average of the given data (values)

Un grouped data

- Arithmetic mean $\bar{x}$ of a set of $x_1, x_2, x_3, \dots\dots\dots x_n$ observations is their sum divided by the number of observations. i.e.
$\bar{x} = \dfrac{x_1 + x_2 + x_3 + \dots\dots \dots + x_n}{N}$
Where $N = 1 + 1 + 1 + \dots\dots \dots + 1 = n$

- If f_i is the frequency of the observations $x_i = 1, 2, \dots\dots\dots, n$ then
$\bar{x} = \dfrac{f_1 x_1 + f_2 x_2 + \dots\dots \dots + f_n x_n}{f_1 + f_2 + \dots\dots \dots + f_n}$

Mean Cont...

- $\bar{x} = \dfrac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i} = \dfrac{\sum fx}{\sum f} = \dfrac{\sum fx}{N}$

Example

Find mean of the set of numbers 63, 65, 67, 68, 69, 70, 71, 72, 73, 74, 75.

Solution

$$\bar{x} = \frac{63+65+67+68+69+70+71+72+73+74+75}{11}$$

$$= 69.73$$

Mean cont...

Example

Calculate the arithmetic mean of the marks from the following table.

Marks	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	12	18	27	20	17	6

Solution

Marks	Class mark (x)	Frequency (f)	*fx*
0-10	5	12	60
10-20	15	18	270
20-30	25	27	675
30-40	35	20	700
40-50	45	17	495
50-60	55	6	330
		$N = \sum f = 100$	$\sum fx = 2800$

$$\bar{x} = \frac{\sum fx}{N} = \frac{2800}{100}$$
$$= 28$$

Mean cont...

ASSUMED MEAN METHOD

- Suppose we have a set of observations $x_1, x_2, \ldots, x_n$ with their frequencies $f_1, f_2, \ldots, f_n$
- Let $\bar{x_a} =$Assumed mean
- Let $d_i = x_i - \bar{x_a}$ (deviation of observation x_i from $\bar{x_a}$) where $i = 1, 2, \ldots n$

- Now, $\sum_{i=1}^{n} f_i d_i = \sum_{i=1}^{n} (x_i - \bar{x_a})$ $= \sum f_i x_i - \bar{x_a} \sum f_i$
- Suppose we divide both sides by($N = \sum f$) we get
- $\dfrac{\sum fd}{N} = \dfrac{\sum fx}{N} = \dfrac{\bar{x_a} \sum f}{N}$
- $\bar{d} = \bar{x} - \bar{x_a}$, Thus $\bar{x} = \bar{d} + \bar{x_a}$,

Mean cont...

Example

- Choose the central value to be the assumed mean.
- Find the mean of the set of numbers
  63, 65, 67, 68, 69, 70, 71, 72, 73, 74, 75.
  by assumed mean method.
- Solution
  Let take $\bar{x_a} = 70$ as the assumed mean, then we have

Numbers(x)	63	65	67	68	69	70	71	72	73	74	75
$d_i = x_i - 70$	-7	-5	-3	-2	-1	0	1	2	3	4	5

$\bar{d} = \frac{\sum d_i}{N} = -0.27$

$\bar{x} = \bar{d} + \bar{x_a} = -0.27 + 70 = 69.73$

Therefore $\bar{x} = 69.73$

Example

- Given the table, find arithmetic mean by
  (a) direct method
  (b) assumed mean method

Class interval	Frequency
0-8	8
8 -16	7
16-24	16
24-32	24
32-40	15
40-48	7

Example

- The following data set represents the age distribution of a sample of 72 women having multiple delivery births in 2018.

Age(Years)	15-19	20-24	25-29	30-34	35-39	40-44
Numbers	2	5	17	28	17	3

  - Find the mean age of the women in years.
    (a) By direct method (b) By using Assumed mean method

Mean cont...

COMBINED MEAN

- Let n_1 =Number of observation of group one
- n_2 = Number of observation of group two
- $\bar{x_1}$ =Arithmetic mean of group one
- $\bar{x_2}$ =Arithmetic mean of group one

Then, the combined mean is given by

- $\bar{x} = \dfrac{n_1\bar{x_1} + n_2\bar{x_2}}{n_1 + n_2}$

Mean cont...

proof

- Let$x_{11}, x_{12}, x_{13}, \ldots, x_{1n} =$ be the observation of the group one
- Let$x_{21}, x_{22}, x_{23}, \ldots, x_{2n} =$ be the observation of the group two
- $\bar{x}_1 = \dfrac{f_{11}x_{11} + f_{12}x_{12} + f_{13}x_{13} + \cdots + f_{1n}x_{1n}}{f_{11} + f_{12} +_{f13} + \cdots + f_{1n}}$ or
- $f_{11}x_{11} + f_{12}x_{12} + f_{13}x_{13} + \cdots + f_{1n}x_{1n} = n_1\bar{x}_1 \ldots\ldots\ldots(1)$
- $\bar{x}_2 = \dfrac{f_{21}x_{21} + f_{22}x_{22} + f_{23}x_{23} + \cdots + f_{2n}x_{2n}}{f_{21} + f_{22} + f_{23} + \cdots + f_{2n}}$ or
- $f_{21}x_{21} + f_{22}x_{22} + f_{23}x_{23} + \cdots + f_{2n}x_{2n} = n_2\bar{x}_2 \ldots\ldots\ldots(2)$
- combined mean
  $\bar{x} = \dfrac{f_{11}x_{11} + f_{21}x_{21} + f_{12}x_{12} + f_{22}x_{22} + \cdots + f_{1n}x_{1n} + f_{2n}x_{2n}}{f_{11} + f_{21} + f_{12} + f_{22} + \cdots + f_{1n} + f_{2n}} \ldots(3)$

Mean cont...

- Using (1) and (2) in (3) Results to;
- $\bar{x} = \dfrac{n_1\bar{x_1} + n_2\bar{x_2}}{n_1 + n_2}$
- in general $\bar{x} = \dfrac{n_1\bar{x_1} + n_2\bar{x_2} + \cdots + n_n\bar{x_n}}{n_1 + n_2 + \cdots + n_n}$

Example

- The mean salary for a group of 40 female workers is $ 5200 per month and for a group of 60 male workers is $ 6800 per month. What is the combined mean salary.

Mean cont...

Solution

- Let n_1 =Number of female workers
- $n_2 =$ Number of male workers
- $\bar{x_1}$ =Mean salary for female workers
- $\bar{x_2}$ =Mean salary for male workers

Then, the combined mean is given by

- $$\bar{x} = \frac{n_1\bar{x_1} + n_2\bar{x_2}}{n_1 + n_2}$$
  $$= \frac{40(5200) + 60(6800)}{40 + 60}$$
  $$= 6160$$
- The combined mean salary is \$ 6160

Mean cont...

Example

- The arithmetic mean of 100 items were recorded as 40, but later on it was discovered that one observation 40 was wrong copied down as 50. Find the correct mean.

Solution
Given:
Number of items = 100.
Incorrect mean = 40.
Incorrect value = 50.
Correct value = 40.
Required the correct mean.

- correct mean$= \dfrac{correct \sum x}{n}$
- but, $correct \sum x =$Incorrect $\sum x$-incorrect value+correct value
- = n(incorrect $\bar{x}$)-incorrect value +correct value.

Mean cont...

- 3990
- Now, the correct mean$= \dfrac{3990}{100} = 39.9$
- Therefore , the correct mean=39.9

Example

Mode

- Mode is a value which occurs most frequently in the given distribution.

Mode for ungrouped data

- Mode for ungrouped data is obtained by checking or inspecting data which occurs mostly.

Mode for grouped data

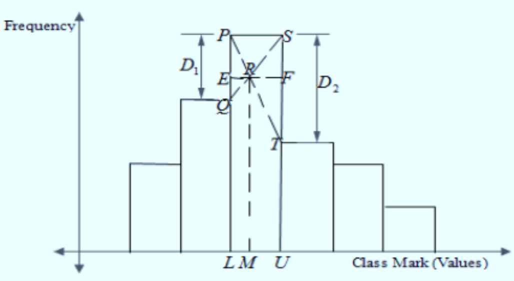

Figure 4: Consider the histogram below

Mode cont...

- From the histogram above
- ΔPRQ is similar to ΔSRT
- $\dfrac{ER}{FR} = \dfrac{PQ}{ST}$
- $\dfrac{M-L}{U-M} = \dfrac{D_1}{D_2}$
- $(M-L)D_2 = (U-M)D_1$
- $MD_2 - LD_2 = UD_1 - MD_1$
- $MD_2 + MD_1 = UD_1 + LD_2$
- $M(D_1 + D_2) = UD_1 + LD_2 \ldots (i)$
- But, class size $C = U - L, U = C + L \ldots$ (ii)
- substitute equation (ii) into (i)

Mode cont...

- $M(D_1 + D_2) = (C + L)D_1 + LD_2$
- $M(D_1 + D_2) = CD_1 + LD_1 + LD_2$
- $M(D_1 + D_2) = +L(D_1 + D_2) + CD_1 \ldots$ (iii)
- divide by $(D_1 + D_2)$ through out equation (iii)
- Mode(M)=$L + \left(\frac{D_1}{D_1 + D_2}\right) C$ where;
- L=Lower class boundary
- D_1=The difference between the frequency of modal class and the frequency of the class before modal class
- D_2= The difference between the frequency of modal class and the frequency of the class after modal class
- C= Class size

Example

- Find the mode of the following scores 40,30,25,10,40,19,40

Solution

- The mode is 40

Example

- Calculate the mean and mode of the following distribution below

Class interval	Frequency
0-30	4
30 -60	7
60-90	14
90-120	11
120-150	2

Solution

- Mean

Class interval	Frequency (f)	x	fx
0-30	4	15	60
30 -60	7	45	315
60-90	14	75	1050
90-120	11	105	1155
120-150	2	135	270

- $\sum f = 38$
- $\sum fx = 2850$
- Mean $(\bar{x})=\dfrac{\sum fx}{\sum f} = \dfrac{2850}{38} = 75$
- Modal class 60-90, L=60, C=30, D_1=14-7=7 and D_2=14-11=3
- Mode (M)=$L + \left(\dfrac{D_1}{D_1 + D_2}\right) C = 60 + (\dfrac{7}{7+3})30 = 81$

Median

Median

- Median is the middle value of the given distribution when arranged in ascending or descending order.

Median from ungrouped data

- Case I: When the number of observations (n) is odd, then median value is $(\frac{n+1}{2})^{th}$ observation.
- Case II: When the number of observations (n) is even, the median is the evarage of the two numbers $(\frac{n}{2})^{th}$ and $(\frac{n}{2}+1)^{th}$ observation.

Median cont...

Median cont...

- Median for grouped data is given by the formula
- Median$= L + \left(\dfrac{\frac{N}{2} - \sum f_b}{f_w} \right) c$

Where,
L= Lower class boundary a of median class
N=Sum of the frequencies
$\sum f_b =$ Sum of frequency before the frequency of median class
$f_w =$ Frequency within the median class
c=Class size

Example

- Find the median of the following scores 19,27,19,18,28,35

Median cont...

Example

- Find the median of the value.
  (i) 25, 20, 15, 14, 18.
  (ii) 16, 17, 20, 12, 30, 38.
- Given the following distribution table,

Class interval	Frequency
21-25	5
26-30	4
31-35	12
36-40	6
41-45	3
46-50	8

Calculate
(a) Mean by assumed method
(b) Mode (c) Median

Solution

Class interval	f	c.f	x	$d = x - \bar{x}_a$	fd
21-25	5	5	23	-10	-50
26-30	4	9	28	-5	-20
31-35	12	21	33	0	0
36-40	6	27	38	5	30
41-45	3	30	43	10	30
46-50	8	38	48	15	120
	$\sum f = 38$				110

- $\bar{x}_a = 33$
- c=5
- $\sum fd = 110$
- $\sum f = 38$
- $\bar{d} = \dfrac{\sum fd}{\sum f} = \dfrac{110}{38}$

  (a) Therefore, Mean $\bar{x} = \bar{x}_a + \bar{d} = 33 + 2.89 = 35.89$

(b) Mode (M)= (M)=$L+\left(\frac{D_1}{D_1+D_2}\right)C$

- The modal class is 31-35
- L=31-0.5=30.5
- D_1=12-4=8
- D_2=12-6=6
- c=5
- Mode (M)= =$L+\left(\frac{D_1}{D_1+D_2}\right)C=30.5+\left(\frac{8}{8+6}\right)5=33.36$

(c)Median

- Median$= L + \left(\dfrac{\frac{N}{2} - \sum f_b}{f_w} \right) c$

  Where,
- L= 30.5, median class 31-35
- N=38
- $\sum f_b = 9$
- $f_w = 12$
- c=5
- Therefore, Median$= L + \left(\dfrac{\frac{N}{2} - \sum f_b}{f_w} \right) c$
- $= 30.5 + \left(\dfrac{\frac{38}{2} - 9}{12} \right) 5 = 34.67$

quetion one

- Find the mode and median of the following distribution below,

Class interval	Frequency
10-20	3
20-30	2
30-40	10
40-50	4
50-60	5
60-70	2

quetion Two

- Find the mode and median of the following distribution below,

Class interval	Frequency
4.0-4.2	3
4.3-4.5	8
4.6-4.8	11
4.9-5.1	13
5.2-5.4	12
5.5-5.7	5
5.8-6.0	1

Question three

- From the following distribution

Class interval	Frequency
0.20-0.24	3
0.25-0.29	10
0.30-0.34	12
0.35-0.39	9
0.40-0.44	2
0.55-0.57	5
0.58-0.60	1

find,
(a) Mean
(b) Mode
(c) Median

Question four

- The mean of two samples of sizes 20 and 42 respectively are 24 and 36. Find the combined mean for the two samples

Question five

- The data below are the lengths of 30 plant leaves measured to nearest mm.construct a frequency distribution table having five classes with class interval 10-14, 15-19 etc.
  10,20,24,25,30,12,20,21,26,31,13,20,22,27,16,
  21,23,28,17,22,22,29,16,21,28,18,17,23,19,15.
  Hence,
  (a) Calculate the mean;
  i) By direct method
  ii) By assumed mean method
  (b) Mode
  (c) Median

MEASURES OF DISPERSION

- This is the measure of spreading (scatter) of the data. These are determinant on how the given data varies from one another or from the mean.
- Measures of dispersion includes;
  (a) Range
  (b) Variance
  (c) Standard deviation

Range

- Range is the difference between greatest and smallest observations or
- Range is the difference between maximum value and minimum value of data.
- Range= Maximum value - Minimum value
- Example: Consider the marks: 30%, 70%, 50%, and 90%.
- Range : greatest observation-smallest observation Therefore 90%-30%= 60%.

Variance

- This is average of squared deviations of individual values from their arithmetic mean.
- Variance of data is denoted by Var(x) or $(\delta)^2$
- Variance for ungrouped data is given by Var(x)=$\dfrac{\sum(x-\bar{x})^2}{n}$
- where,
  x=Individual data
  $\bar{x}$= Mean
  n=Number of observations

variance cont...

- Var(x)=$\dfrac{\sum(x-\bar{x})^2}{n}$
- Var(x)=$\dfrac{\sum(x^2-2x\bar{x}+\bar{x}^2)}{n}$
- Var(x)=$\dfrac{\sum x^2-\sum 2x\bar{x}+\sum \bar{x}^2}{n}$
- Var(x)=$\dfrac{\sum x^2}{n}-\dfrac{2\bar{x}\sum x}{n}+\bar{x}^2$
- Var(x)=$\dfrac{\sum x^2}{n}-2\bar{x}^2+\bar{x}^2$
- Var(x)=$\dfrac{\sum(x-\bar{x})^2}{n}=\dfrac{\sum x^2}{n}-\bar{x}^2$

Example

Given the following, n=10, $\sum x = 60$ and $\sum x^2 = 1000$, find
(a) mean
(b) Variance
Solution
(a) mean($\bar{x}$)=$\dfrac{\sum x}{n} = \dfrac{60}{10} = 10$
(b) Variance=$\dfrac{\sum x^2}{n} - \bar{x}^2 = \dfrac{1000}{10} - 6^2 = 64$

Example

It is given that N=200, $\bar{x}$=48 and$\delta = 3$ its required to find the values of $\sum x$ and $\sum x^2$

Solution

Recall, from the formula

$\bar{x} = \dfrac{\sum x}{n} = \sum x = N * \bar{x}$

$\sum x = 200 * 48 = 9600$ From Variance var(x)=$(\delta)^2 = \dfrac{\sum x^2}{N} - \bar{x}^2$

$\sum x^2 = N * (\delta)^2 + N * \bar{x}^2$

$= 200 * 3^2 + 200 * 48^2 = 1800 + 460800 = 462600$

Example

The mean and variance of 7 observations are 8 and 16 respectively. If five of the observations are 2,10, 12 and 14. Find the remaining two observations.

Example

The mean and variance of 6 observations are 8 and 4 respectively. If each observation is multiplied by 3, find the new mean and new variance of the resulting observations

Example

The mean and standard deviation of 20 observations are found to be 10 and 2 respectively. On rechecking, it was found that an observation 8 was incorrect. Calculate the correct mean and standard deviation in each of the following cases,
(a) If the wrong item was omitted.
(b) If it is replaced by 12.

Variance for Grouped data

- Variance by direct method
- Variance for grouped data by direct method is given by

  Var(x)=$\dfrac{\sum f(x-\bar{x})^2}{\sum f}$

  Where;

  f=frequency

  x=class mark

  $\bar{x}$=mean

  $\sum f$= sum of frequency

Variance for Grouped data cont...

- From the above formula we can express the variance formula in another way as shown below,
- Var(x)=$\dfrac{\sum f(x-\bar{x})^2}{\sum f}$
- Var(x)=$\dfrac{\sum f(x^2-2x\bar{x}+\bar{x}^2)}{\sum f}$
- Var(x)=$\dfrac{\sum fx^2-2\sum fx\bar{x}+\sum f\bar{x}^2}{\sum f}$
- Var(x)=$\dfrac{\sum fx^2-2\bar{x}\sum fx+\bar{x}^2\sum f}{\sum f}$
- Var(x)=$\dfrac{\sum fx^2}{\sum f}-\dfrac{2\bar{x}\sum fx}{\sum f}+\dfrac{\bar{x}^2\sum f}{\sum f}$ but $\dfrac{\sum fx}{\sum f}=\bar{x}$
- Var(x)=$\dfrac{\sum fx^2}{\sum f}-2\bar{x}^2+\bar{x}^2=\dfrac{\sum fx^2}{\sum f}-\bar{x}^2$ or Var(x)=$\dfrac{\sum fx^2}{\sum f}-\left(\dfrac{\sum fx}{\sum f}\right)^2$

Variance for Grouped data cont...

- Variance by assumed method,
- Var(x)=$\frac{\sum f(x-\bar{x})^2}{\sum f}$ ... (i)
- Var(x)=$\frac{\sum f[(x-A)+(A-\bar{x})]^2}{\sum f}$
- Var(x)=$\frac{\sum f[(x-A)^2-2(x-A)(A-\bar{x})+(A-\bar{x})^2]}{\sum f}$
- Var(x)=$\frac{\sum f(x-A)^2-2\sum f(x-A)(A-\bar{x})+\sum f(A-\bar{x})^2}{\sum f}$
- Var(x)=$\frac{\sum f(x-A)^2}{\sum f}-2(A-\bar{x})\frac{\sum f(x-A)}{\sum f}+\frac{(A-\bar{x})^2)\sum f}{\sum f}$
- Var(x)=$\frac{\sum f(x-A)^2}{\sum f}+2(A-\bar{x})\frac{\sum f(x-A)}{\sum f}+(A-\bar{x})^2 \ldots$ (ii)
- but $\bar{x}=A+\frac{\sum fd}{\sum f}$, where $d=x-A$, $A-\bar{x}=-\frac{\sum f(x-A)}{\sum f}$ ... (iii)

Variance for Grouped data cont...

- Substitute equation (iii) into equation (ii),
- Var(x)=$\frac{\sum f(x-A)^2}{\sum f} - 2\frac{\sum f(x-A)}{\sum f}\frac{\sum f(x-A)}{\sum f} + \left(\frac{\sum f(x-A)}{\sum f}\right)^2$
- Var(x)=$\frac{\sum f(x-A)^2}{\sum f} - 2\left(\frac{\sum f(x-A)}{\sum f}\right)^2 + \left(\frac{\sum f(x-A)}{\sum f}\right)^2$
- Var(x)=$\frac{\sum f(x-A)^2}{\sum f} - \left(\frac{\sum f(x-A)}{\sum f}\right)^2$
- but x-A=d, Therefore.
- Var(x)=$\frac{\sum fd^2}{\sum f} - \left(\frac{\sum fd}{\sum f}\right)^2$

Standard deviation

- Standard deviation is a measure that is used to quantify the amount of variation or dispersion of a set of data values,
- Is the measure of dispersion of a set of data from its mean. item Standard deviation is denoted by a Greek letter δ
- Standard deviation is given by taken the square root of Variance.
- Standard Deviation (S.D)=$\sqrt{Var(X)}$

SIGNIFICANCE OF STANDARD DEVIATION

- If the standard deviation is zero, it indicates that there is no deviation and all observations are equal to mean.
- If the standard deviation is small, it indicates that the observations are close to the mean.
- If the standard deviation is large, it indicates that there is high degree of dispersion of observations from the mean

Standard deviation for ungrouped data

- Standard deviation(S.D)=$\sqrt{Var(x)} = \sqrt{\frac{\sum(x-\bar{x})^2}{N}}$

Standard deviation for grouped data

- Standard deviation(S.D) by direct method

S.D=$\sqrt{Var(x)} = \sqrt{\frac{\sum f(x-\bar{x})^2}{\sum f}}$

- Standard deviation(S.D) by Assumed mean method

S.D=$\sqrt{Var(x)} = \sqrt{\frac{\sum fd^2}{\sum f} - \left(\frac{\sum fd}{\sum f}\right)^2}$

Example

Given the distribution table below;

Class Interval	Frequency
1-5	5
6-10	10
11-15	12
16-20	6
21-25	8

Calculate Variance and standard deviation;

(a) By direct method

(b) By assumed method

Solution

Class I.	x	f	c.f	fx	fx^2	d=x-A	fd^2
1-5	3	5	5	15	45	-10	-50
6-10	8	10	15	80	640	-5	-40
11-15	13	12	27	156	2028	0	0
16-20	18	6	33	108	1988	5	30
21-25	23	8	41	184	4232	10	80

A=13, C=5

$\sum f$=41

$\sum fx$=543

$\sum fx^2$=8889

$\sum fd$ =10

$\sum fd^2$=1700

$\sum fU$ =2

$\sum fU^2$ =68

Solution

(a) Variance and standard deviation by direct method

$$\text{var(x)}=\frac{\sum fx^2}{\sum f}-\left(\frac{\sum fx}{\sum f}\right)=\frac{1889}{41}-\left(\frac{543}{41}\right)^2=41.4$$

Standard Deviation$=\sqrt{Var(x)}=\sqrt{41.4}=6.4$

(b)Variance and standard by assumed method

$$\text{var(x)}=\frac{\sum fd^2}{\sum f}-\left(\frac{\sum fd}{\sum f}\right)=\frac{1700}{41}-\left(\frac{10}{41}\right)^2=41.4$$

Standard Deviation$=\sqrt{Var(x)}=\sqrt{41.4}=6.4$

Example

The means of two samples of sizes 50 and 100 respectively are 54.1 and 50.3. Find combined mean for the two samples.

example

The means of two samples of sizes 20 and 42 respectively are 24 and 36. Find combined mean for the two samples.

example

The means of two samples of sizes 18 and 26 respectively are 21 and 23. Find combined mean for the two samples.

MEAN AND VARIANCE OF THE COMBINED SAMPLES

Consider the following two samples of observations where,
(i) n_1 and n_2 are number of observations of two samples respectively.
(ii) $\bar{x_1}$ and$\bar{x_2}$ are means of two samples respectively.
(iii) are standard deviations of two samples respectively.
(iv) d_1 and d_2 are the difference of two mean from combine mean respectively.

The Variance of the combined samples is given by the formula below

Combined Variance V(x)=$\dfrac{n_1\left(\delta_1^2+d_1^2\right)+n_2\left(\delta_2^2+d_2^2\right)}{n_1+n_2}$

example

The mean of two samples of sizes 50 and 100 respectively are 54.1 and 50.3 and the standard deviation are 8 and 7. Find combined mean and the standard deviation of the two samples.

Solution

Given $n_1 = 50$,$\bar{x_1} = 54.1$,$\delta_1 = 8$

$n_2 = 100$,$\bar{x_2} = 5 - 0.3$, $\delta_2 = 7$

From Combined mean formula

$$\bar{x} = \frac{n_1\bar{x_1} + n_2\bar{x_2}}{n_1 + n_2} = \frac{(50 * 54.1) + (100 * 50.3)}{50 + 100} = 51.57$$

From Combined Variance, $\delta^2 = \dfrac{n_1(\delta_1^2 + d_1^2) + n_2(\delta_2^2 + d_2^2)}{n_1 + n_2}$

$d_1 = \bar{x_1} - \bar{x} = 54.1 - 51.57 = 2.53$

$d_2 = \bar{x_2} - \bar{x} = 50.3 - 51.57 = -1.27$

$$\delta^2 = \frac{50(8^2 + 2.53^2) + 100(7^2 + 1.27^2)}{50 + 100} = 57.2089$$

$\delta = \sqrt{57.2089} = 7.56$ The combined standard deviation is 7.56

example

The mean and standard deviation of a group of 100 observations were found to be 20 and 3 respectively, later on it was found that three observations were incorrect, which are recorded as 21, 21 and 18. Find the mean and standard deviation if the incorrect observations were omitted.

example

Salaries paid to supervisors had a mean of 2500/=with the standard deviation of 2000/=. If all salaries are increased by 2500/= , find the new mean and standard deviation.

example

The mean and standard deviation of a set of 100 items were found to be 60 and 10, respectively. At the time of calculation, two items were wrongly taken as 5 and 45 instead of 30 and 20, respectively. Find the corrected standard deviation

example

If the mean of 5 observations is 4.4 and their variance is 8.24, find the values of the two observations given the three observation are 1, 2 and 6.

example

The following table represents the heights taken to the nearest centimetres of 40 orange trees in the garden:

Height (cm)	Frequency
131-140	3
141-150	4
151-160	k
161-170	11
171-180	k+2
181-190	5
191-200	k-6

(a) Find the value of k
(b) By using assumed method determine
(i) The actual mean height
(ii) The standard deviation of the distribution

COEFFICIENT OF VARIATION

- The Standard deviation is an absolute measure of dispersion. It is expressed in terms of units in which the original figures are collected and stated.
- The standard deviation of heights of students cannot be compared with the standard deviation of weights of students, as both are expressed in different units, i.e heights in centimetres and weights in kilograms.
- Therefore the standard deviation must be converted into a relative measure of dispersion for the purpose of comparison.
- The relative measure is known as the coefficient of variation.
- The coefficient of variation is obtained by dividing the standard deviation by the mean and multiply it by 100%. symbolically,

C. V cont....

Coefficient of variation (C.V) $= \frac{s}{\bar{x}} \times 100\%, \quad \text{or} \quad C.V = \frac{\sigma}{\bar{\mu}} \times 100\%.$

- If we want to compare the variability of two or more series, we can use C.V.
- The groups of data for which the **C.V. is greater** indicate that the group is more variable, less stable, less uniform, less consistent or less homogeneous.
- If the **C.V. is less**, it indicates that the group is less variable, more stable, more uniform, more consistent or more homogeneous.

c.v cont....

Example

In two factories A and B located in the same industrial area, the average weekly wages (in rupees) and the standard deviations are as follows

Factory	Average	Standard Deviation	No. of workers
A	34.5	5	476
B	28.5	4.5	524

i. Which factory A or B pays out a larger amount as weekly wages?

ii. Which factory A or B has greater variability in individual wages?

Solution

- Given Factory A, $N_1 = 476, \quad \bar{x}_1 = 34.5, \quad \sigma_1 = 5$
  Factory B, $N_2 = 524, \quad \bar{x}_2 = 28.5, \quad \sigma_2 = 4.5$
- i. Total wages paid by factory A; $= 34.5 \times 476 = \text{Rs.}16,422$.
  Total wages paid by factory B $= 28.5 \times 524 = \text{Rs.}14,934$.
  Therefore factory A pays out larger amount as weekly wages.
- ii. C.V. of distribution of weekly wages of factory A and B are

$$\text{Coefficient of variation } (\text{C.V})_A = \frac{\sigma_1}{\bar{x}_1} \times 100\% = \frac{5}{34.5} \times 100\% = 14.49\%$$

$$\text{Coefficient of variation } (\text{C.V})_B = \frac{\sigma_2}{\bar{x}_2} \times 100\% = \frac{4.5}{28.5} \times 100\% = 15.79\%$$

- Therefore Factory B has greater variability in individual wages, since C.V. of factory B is greater than C.V of factory A.

MEASURES OF POSITION

- Measures of position includes:
  (a) Quartile
  (b) Percentile

QUARTILE

- Quartile is the division of frequency distribution into four equal parts, or quartile are data values which divides the distribution into quarters.
- Let N be number of observations of the frequency distribution.

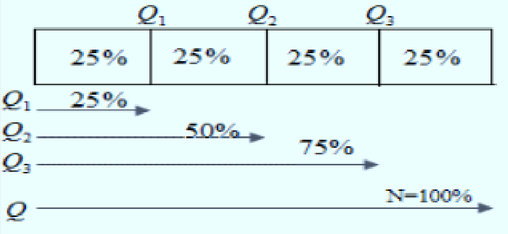

Figure 5: Partition values

QUARTILE Cont...

Positions of Quartile for Ungrouped data are divided into three types

- Position of First Quartile (Lower Quartile)
  (a) Position of Lower Quartile given even number of observations is $(\frac{1}{4}N)^{th}$
  (b) Position of Lower Quartile given odd number of observations is $\left(\frac{1}{4}(N+1)\right)^{th}$

QUARTILE Cont...

Positions of Quartile for Ungrouped data are divided into three types

- Position of second Quartile (Median)
  (a) Position of second Quartile given even number of observations is $(\frac{2}{4}N)^{th}$
  (b) Position of second Quartile given odd number of observations is $\left(\frac{2}{4}(N+1)\right)^{th}$

QUARTILE Cont...

Positions of Quartile for Ungrouped data are divided into three types

- Position of Third Quartile (Upper Quartile)
  (a) Position of third Quartile given even number of observations is $(\frac{3}{4}N)^{th}$
  (b) Position of third Quartile given odd number of observations is $\left(\frac{3}{4}(N+1)\right)^{th}$

Interquartile Range (I.Q.R)

- This is the difference between upper quartile and lower quartile
- I.Q.R=Upper Quartile - Lower Quartile
- $I.Q.R = Q_3 - Q_1$

Semi-Interquartile Range (S. I.R)

- Semi-Interquartile Range (S.I.R) is given by,
- Semi I.Q.R=(Upper Quartile - Lower Quartile)/2
- Semi I.Q.R=$\dfrac{Q_3 - Q_1}{2}$
- **Note:** Semi-Interquartile range sometimes is called Semi-InterQuartile Deviation

QUARTILE Cont...

Positions of Quartile for Grouped data are divided into three types

- Lower Quartile
$$Q_1 = L + \left(\frac{\frac{N}{4} - f_b}{f_w}\right)c$$
- Second Quartile (Median)
$$Q_2 = L + \left(\frac{\frac{N}{2} - f_b}{f_w}\right)c$$
- Third Quartile
$$Q_3 = L + \left(\frac{\frac{3N}{4} - f_b}{f_w}\right)c$$

Percentile

Percentile is the division of frequency distribution into 100 equal parts as shown below,

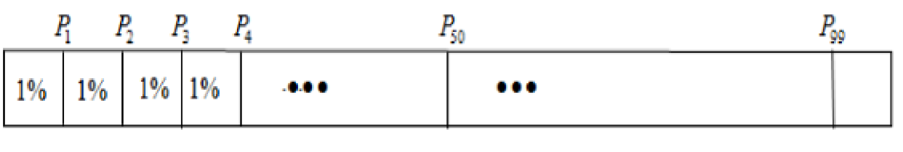

Figure 6: Percentile values

Percentile for UnGrouped Data

- The position of 1^{st} percentile is $(\frac{N}{100})^{th}$
- The position of 2^{nd} percentile is $(\frac{2N}{100})^{th}$
- The position of 3^{rd} percentile is $(\frac{3N}{100})^{th}$

Percentile for UnGrouped Data

- The position of 4^{th} percentile is $(\frac{4N}{100})^{th}$
- ...
- The position of 50^{th} percentile is $(\frac{50N}{100})^{th}$
- ...
- The position of 99^{th} percentile is $(\frac{99N}{100})^{th}$

Percentile for Grouped Data

- The position of 1^{st} percentile is given by $P_1 = L + \left(\dfrac{\frac{N}{100} - f_b}{f_w}\right)c$
- The position of 2^{th} percentile is given by $P_2 = L + \left(\dfrac{\frac{2N}{100} - f_b}{f_w}\right)c$

Percentile for Grouped Data

- ...
- The position of 50^{th} percentile is given by $P_{50} = L + \left(\frac{\frac{50N}{100} - f_b}{f_w} \right) c$
- ...
- The position of 99^{th} percentile is given by $P_{99} = L + \left(\frac{\frac{99N}{100} - f_b}{f_w} \right) c$
- Where,
  L is the lower limit of the n^{th} quartile class or percentile class
  N is the total number of observations.
  $\sum f_b$ is the cumulative frequency of the previous class to the n^{th} quartile class or percentile class
  f_w is the frequency of the n^{th} quartile class or percentile class.
  c is the class size.

Percentile range

- This is a difference between two specified percentiles.
- These could theoretically be any two percentiles, but the$10^{th} - 90^{th}$ percentile range is most common.

Example

Find the lower quartile, upper quartile and IQR of 9,19,27,14 ,28,29,39.

Solution

- In ascending order 9,14,19,27,28,29,39.
- N=7
- Lower Quartile $Q_1 = \left(\frac{1}{4}(N+1)\right)^{th} = \left(\frac{1}{4}(7+1)\right)^{th} = 2^{nd} = 14$
- Upper Quartile $Q_3 = \left(\frac{3}{4}(N+1)\right)^{th} = \left(\frac{3}{4}(7+1)\right)^{th} = 6^{th} = 29$
- Inter Quartile Range (I.Q.R)= $Q_3 - Q_1 = 29 - 14 = 15$

example

Find the quartile deviation of the following distributions 7,10,11,12,13,14,16,17

Solution

- In ascending order 7,10,11,12,13,14,16,17.
- N=8
- Lower Quartile $Q_1 = \left(\frac{1}{4}N\right)^{th} = \left(\frac{1}{4}8\right)^{th} = 2^{nd} = 10$
- Upper Quartile $Q_3 = \left(\frac{3}{4}N\right)^{th} = \left(\frac{3}{4}8\right)^{th} = 6^{th} = 14$
- Inter Quartile Range (I.Q.R)= $Q_3 - Q_1 = 14 - 10 = 4$
- Semi- InterQuartile Range (s. I.R)= $\frac{Q_3 - Q_1}{2} = \frac{14 - 10}{2} = 2$

example

The following table shows distribution of marks on a final examination in Advanced Mathematics.

Marks	Number of students
30-39	1
40 -49	3
50-59	11
60-69	21
70-79	43
80-89	32
90-99	9

Find;

(a) Find the quartiles of the distribution

(b) Find the quartile deviation

(c) Semi-interquartile deviation

(d) Find the class interval with 10^{th} and 90^{th} percentiles

(e) Percentile range

example

Solution

Marks	Number of students	c.f
30-39	1	1
40 -49	3	4
50-59	11	15
60-69	21	36
70-79	43	79
80-89	32	111
90-99	9	120

(a) Lower quartiles $Q_1 = L + \left(\dfrac{\frac{N}{4} - f_b}{f_w} \right) c$

Position of $Q_1 = (\frac{N}{4})^{th} = (\frac{120}{4})^{th} = 30^{th}$ its class interval 60-69

L=59.5

f_b=15
f_w=21
c=10

then $Q_1 = L + \left(\dfrac{\frac{N}{4} - f_b}{f_w} \right) c = 59.5 + \left(\dfrac{\frac{120}{4} - 15}{21} \right) 10 = 66.6429$

Upper quartiles $Q_3 = L + \left(\dfrac{\frac{3N}{4} - f_b}{f_w} \right) c$

Position of $Q_3 = (\frac{3N}{4})^{th} = (\frac{3*120}{4})^{th} = 90^{th}$ its class interval 80-89

L=79.5
f_b=79
f_w=32
c=10

then $Q_3 = L + \left(\dfrac{\frac{3N}{4} - f_b}{f_w} \right) c = 79.5 + \left(\dfrac{\frac{3*120}{4} - 15}{21} \right) 10 = 82.9375$

(b) quartile deviation=$Q_3 - Q_1 = 82,9375 - 66.6429 = 16.2946$

(c) Semi-quartile deviation=$\dfrac{Q_3 - Q_1}{2} = \dfrac{16.2946}{2} = 8.1473$

(d) class interval with 10^{th} and 90^{th} percentiles

Position of 10^{th} percentile=$(\dfrac{10N}{100})^{th} = (\dfrac{10 * 120}{100})^{th} = 12^{th}$ its class interval 50-59

Position of 90^{th} percentile=$(\dfrac{90N}{100})^{th} = (\dfrac{90 * 120}{100})^{th} = 108^{th}$ its class interval 80-89

(e) Percentile range=$90^{th} - 10^{th}$

10^{th} percentiles $P_{10} = L + \left(\dfrac{\frac{10N}{100} - f_b}{f_w} \right) c$

Position of $P_{10} = (\dfrac{10N}{100})^{th} = (\dfrac{10 * 120}{100})^{th} = 12^{th}$ its class interval 50-59

L=49.5

f_b=4

f_w=11

c=10

then $P_{10} = L + \left(\dfrac{\frac{10N}{100} - f_b}{f_w} \right) c = 49.5 + \left(\dfrac{\frac{10*120}{100} - 4}{11} \right) 10 = 56.77$

90^{th} percentiles $P_{90} = L + \left(\frac{\frac{90N}{100} - f_b}{f_w} \right) c$

Position of $P_{10} = (\frac{90N}{100})^{th} = (\frac{90*120}{100})^{th} = 108^{th}$ its class interval 80-89

L=79.5

f_b=79

f_w=32

c=10

then $P_{90} = L + \left(\frac{\frac{90N}{100} - f_b}{f_w} \right) c = 79.5 + \left(\frac{\frac{90*120}{100} - 79}{32} \right) 10 = 88.56$

percentile range=$90^{th} - 10^{th} = 88.56 - 56.77 = 31.79$

OUTLIERS

- **Outlier** is an extremely high or an extremely low data value when compared with the rest of the data values.
- When the data set contain an outlier then the Mean is not a good measure of central tendency, in this case we use the median.
- A possible outlier (mild outlier) will be any data point that lies below $Q_1 - 1.5(IQR)$ or
  above $Q_3 + 1.5(IQR)$.

Example

Find Q_1, Q_2, and Q_3 for the data set: 20, 10, 2, 11, 14, 8, 17, 5. Check for the outlier.

Solution

Arranging data in ascending order gives: 2, 5, 8, 10, 11, 14, 17, 20.

Computation for $Q_2 = (10 + 11)/2 = 10.5$

Computation for $Q_1 : 2, 5, 8, 10 \quad \therefore Q_1 = 6.5$

OUTLIERS CONT....

Computation for $Q_3 : 11, 14, 17, 20 \quad \therefore Q_3 = 15.5$
$\therefore Q_1 = 6.5, Q_2 = 10.5 \quad \text{and} \quad Q_3 = 15.5.$

$$\text{IQR} = 15.5 - 6.5 = 9.$$
$$Q_1 - 1.5(IQR) = 6.5 - 1.5(9) = 6.5 - 13.5 = -7.$$
$$Q_3 + 1.5(IQR) = 15.5 + 1.5(9) = 6.5 + 13.5 = 29$$

Check the data set for any data values that fall outside the interval from -7 to 29. There is no data that is below -7 or above 29. Thus there is no outlier.

Example

Determine the median and 1st and 3rd quartile values for this data: 53, 62, 78, 94, 96, 99, 103. Check for the outlier.

Example

Find Q_1, Q_2, and Q_3 for the data set: 34, 20, 10, 16, 17, 28, 12, 30, 38.

OUTLIERS CONT....

- The number of working days lost due to accidents for each of 12 one-monthly periods are as shown . Determine the median and first and third quartile values for this data.
  27 37 40 28 23 30 35 24 30 32 31 28.
  Check for the outliers.
- Check the following data set for outliers: 14, 16, 27, 18, 13, 19, 36, 15, 20.

SKEWNESS

- Skewness is a statistical measure that describe the asymmetry of a probability distribution.
- Skewness denote the opposite of symmetry. it is lack of symmetry. In a symmetrical series the mode, median and the arithmetic mean are identical.
- Skewness is used as a measure of the asymmetry (lack of symmetry) of a density function about its mean.
- If in a distribution **mean = median = mode**, then that distribution is known as symmetrical distribution (normal distribution).

Skewness cont...

- If in a distribution **mean** $\neq$ **median** $\neq$ **mode**, then it is not a symmetrical (asymmetrical) distribution and it is called a skewed distribution and such a distribution could either be positively skewed or negatively skewed.
- A negative skewed distributionhas a tail on the left and a peak on the right.
- A positive skewed distribution has a tail on the right and a peak on the left.
- Skewness helps to assess the departure from symmetry, bell-shaped distribution.

Symmetrical distribution

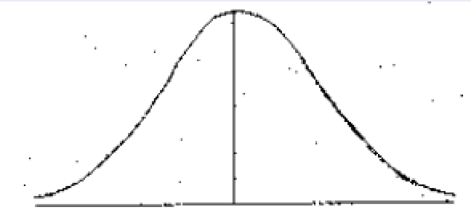

Figure 7: $mean = mode = median$

Positive skewed distribution

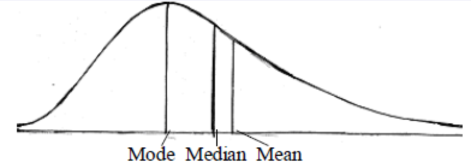

Figure 8: $Mode < Mdian < Mean$

Negative skewed distribution

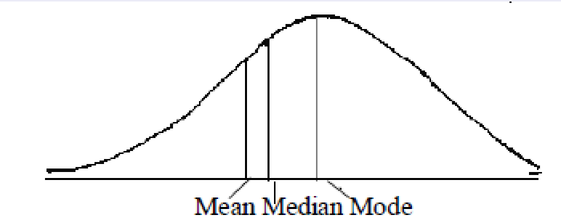

Figure 9: $mean < median < mode$

Measures of skewness

The important measures of skewness are:

(i) Karl-Pearson's coefficient of skewness
(ii) Bowley's coefficient of skewness
(iii) Measure of skewness based on moments

The coefficient of skewness

The coefficient of skewness is :

(i) **zero (0)** when data are Symmetric about the mean (normal distribution).

(ii) **greater than 0** when data have Positive skew/longer right tail.

(iii) **less than 0** when data have Negative skew/longer left tail.

Karl-Pearson's Coefficient of skewness

According to **Karl-Pearson**, the absolute measure of skewness = mean - mode. This measure is not suitable for making valid comparison of the skewness in two or more distributions because the unit of measurement may be different in different series. To avoid this difficulty use relative measure of skewness called Karl-Pearson's coefficient of skewness given by:

$$\text{Karl-Pearson's coefficient of skewness} = \frac{\text{mean} - \text{mode}}{\text{standard deviation}}$$

Coefficient of skewness cont..

In case of mode is ill-defined, the coefficient can be determined by the formula:

$$\text{Coefficient of skewness} = \frac{3(\text{mean} - \text{median})}{\text{standard deviation}}$$

Example

Calculate Karl-Pearson's coefficient of skewness for the following sample data.
25, 15, 23, 40, 27, 25, 23, 25, 20.

Solution
arrange the given data in ascending order
15, 20, 23, 23, 25, 25, 25, 27, 40
The mode is 25

coefficient of skewness cont...

x	$d = x - A$	d^2	$x - \bar{x}$	$(x - \bar{x})^2$
15	-10	100	-9.8	96.04
20	-5	25	-4.8	23.04
23	-2	4	-1.8	3.24
23	-2	4	-1.8	3.24
25	0	0	0.2	0.004
25	0	0	0.2	0.004
25	0	0	0.2	0.004
27	2	4	2.2	4.84
40	15	225	15.2	231.04
$n = 9$	$\sum d = -2$	$\sum d^2 = 362$		

The mean is

$$\bar{x} = A + \frac{\sum d}{n} = 25 + \frac{-2}{9} = 24.8$$

coefficient of skewness cont...

The standard deviation is

$$s = \sqrt{\frac{(x-\bar{x})^2}{n-1}} = \sqrt{\frac{361.452}{8}} = \sqrt{45.1815} = 6.7$$

$$\text{Karl-Pearson's coefficient of skewness} = \frac{\text{mean} - \text{mode}}{\text{standard deviation}} = \frac{24.8-25}{6.7}$$

$$\text{Karl-Pearson's coefficient of skewness} = \frac{-0.2}{6.7} = -0.03 \approx 0.$$

Bowley's Coefficient of skewness

- In Karl-Pearson's method of measuring skewness the whole of the series is needed.
- Prof. Bowley has suggested a formula based on relative position of quartiles.
- In a symmetrical distribution, the quartiles are equidistant from the value of the median; that is, Median $-Q_1 = Q_3-$ Median.
- But in a skewed distribution, the quartiles will not be equidistant from the median.
- Hence Bowley has suggested the following formula:

$$\text{Bowley's coefficient of skewness (sk)} = \frac{Q_3 + Q_1 - 2\text{Median}}{Q_3 - Q_1}$$

Bowley's coefficient

Example

Find the Bowley's coefficient of skewness for the following series. 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22.

Solution Median $= Q_2 = 12, \qquad Q_1 = 6, \qquad Q_3 = 18.$

$$\text{Bowley's coefficient of skewness (sk)} = \frac{Q_3 + Q_1 - 2\text{Median}}{Q_3 - Q_1}$$

$$\text{Bowley's coefficient of skewness (sk)} = \frac{18 + 6 - 2(12)}{18 - 6} = 0$$

Since $sk = 0$, the given series is a symmetrical data.

Measure of skewness based on moments

The measure of skewness based on moments is denoted by β_1 and is given by

$$\beta_1 = \frac{\mu_3}{\mu_2^{3/2}}, \qquad \text{If } \mu_3 \text{ is negative, then } \beta_1 \text{ is negative}$$

Example

Calculate the coefficient of skewness β_1 for the following population data.

x	2	4	6	8	10
f	1	4	6	4	1

Solution

x	f	fx	$x_i - \mu$	$f_i(x_i - \mu)$	$f_i(x_i - \mu)^2$	$f_i(x_i - \mu)^3$	$f_i(x_i - \mu)^4$
2	1	2	-4	-4	16	-64	256
4	4	16	-2	-8	16	-32	64
6	6	36	0	0	0	0	0
8	4	32	2	8	16	32	64
10	1	10	4	4	16	64	256
	N=16	96		0	64	0	640

$$\mu = \frac{\sum fx}{N} = \frac{96}{16} = 6$$

solution cont...

Calculating the first four moments about the mean

$$\mu_1 = \frac{f_i(x_i - \mu)}{N} = \frac{0}{16} = 0$$

$$\mu_2 = \frac{\sum f_i(x_i - \mu)^2}{N} = \frac{64}{16} = 4$$

$$\mu_3 = \frac{\sum f_i(x_i - \mu)^3}{N} = \frac{0}{16} = 0$$

$$\mu_4 = \frac{\sum f_i(x_i - \mu)^4}{N} = \frac{640}{16} = 40$$

Using the second moment μ_2 and third moment μ_3 to calculate the β_1

$$\beta_1 = \frac{\mu_3}{\mu_2^{3/2}} = \frac{0}{\sqrt{4^3}} = 0, \qquad \text{It is normal distribution}$$

Kurtosis

- Kurtosis is the measure of the degree of peakedness of a distribution
- Kurtosis is a statisticasl measure that describes the distribution of the data.
- It indicates the tail heavness (leptokurtic) or lightnes (platykurtic) of the distributioncompared to a normal distribution.
- High Kutosis suggest more data in the tail , while low kurtosis indicates less extreme values

Measures of Kurtosis

Measure of Kurtosis is given by $\beta_2 = \dfrac{\mu_4}{\mu_2^2}$

where $\mu_2 = \dfrac{\sum(x-\bar{x})^2}{N}$ and $\mu_4 = \dfrac{\sum(x-\bar{x})^4}{N}$

Measures of Kurtosis

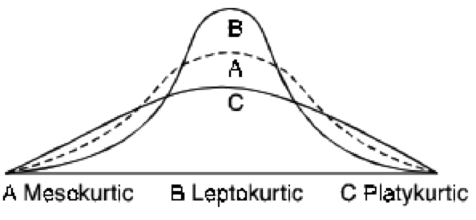

Figure 10: Measures of Kurtosis

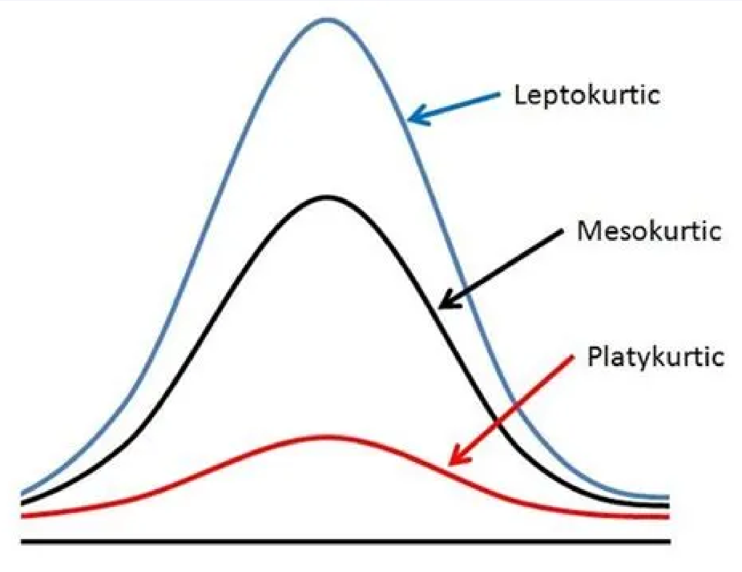

Figure 11: Measures of Kurtosis

- Leptokurtic: A distribution with positive excess kurtosis is called leptokurtic.
- Leptokurtic distributions have a higher peak than a normal distribution and heavier tails, indicating that they have a higher probability of extreme values.
- This means that leptokurtic distributions are riskier than a normal distribution.
- Mesokurtic: A distribution with an excess kurtosis of 0 is called mesokurtic.

- Mesokurtic distributions have the same peakedness as a normal distribution, but may have heavier or lighter tails, depending on the distribution.
- This means that mesokurtic distributions are generally considered to be less risky than leptokurtic distributions.
- Platykurtic: A distribution with negative excess kurtosis is called platykurtic.
- Platykurtic distributions have a lower peak than a normal distribution and lighter tails, indicating that they have a lower probability of extreme values.
- This means that platykurtic distributions are generally considered to be less risky than leptokurtic distributions

example

Calculate sample standard deviation, β_1 and β_2 for the following data.

70	71	73	68	67
73	69	72	76	71

Calculate β_1 and β_2 for the following data.

26	12	16	56	112	24

Example

Calculate β_1 and β_2 for the following data.

x	2	4	6	8	10
f	1	4	6	4	1

Solution

x	f	fx	$x_i - \bar{x}$	$f_i(x_i - \bar{x})$	$f_i(x_i - \bar{x})^2$	$f_i(x_i - \bar{x})^3$	$f_i(x_i - \bar{x})^4$
2	1	2	-4	-4	16	-64	256
4	4	16	-2	-8	16	-32	64
6	6	36	0	0	0	0	0
8	4	32	2	8	16	32	64
10	1	10	4	4	16	64	256
	N=16	96		0	64	0	640

$$\bar{x} = \frac{\sum fx}{N} = \frac{96}{16} = 6$$

$$\mu_2 = \frac{\sum f_i(x_i - \bar{x})^2}{N} = \frac{64}{16} = 4$$

$$\mu_3 = \frac{\sum f_i(x_i - \bar{x})^3}{N} = \frac{0}{16} = 0$$

$$\mu_4 = \frac{\sum f_i(x_i - \bar{x})^4}{N} = \frac{640}{16} = 40$$

$$\beta_1 = \frac{\mu_3}{\mu_2^{3/2}} = \frac{0}{\sqrt{4^3}} = 0, \qquad \text{It is normal distribution}$$

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{40}{4^2} = 2.5, \qquad \text{The curve is platykurtic}$$

Questions

(1). Twenty-six dental patients require the following numbers of fillings during their current course of treatment:

2	3	2	2	3	1	2	2	1	3	2	2	2
2	4	3	2	2	2	2	2	1	1	0	1	1

(a) Identify the mode, find the median, and calculate the mean of these figures.

(b) Compare the results you obtained for (a). What do they tell you about the shape of the distribution?

(c) Find the median, upper and lower quartiles for this distribution.

(2). Particulars regarding the income of two villages are given below:

	Village A	Village B
Number of people	600	500
Average income (in Rs)	175	186
Variance of income (in Rs)	100	81

In which village is the variation in income greater?

(3). From the following table calculate the Karl-Pearson's coefficient of skewness

Daily Wages (in Rs)	150	200	250	300	350	400	450
No. of People	3	25	19	16	4	5	6

(4). Compute Bowley's coefficient of skewness from the following data

Size	5-7	8-10	11-13	14-16	17-19
Frequency	14	24	38	20	4

(5). Using moments calculate β_1 and β_2 from the following data

Daily wages	70-90	90-110	110-130	130-150	150-170
No. of workers	8	11	18	9	4

(6). Two workers on the same job show the following result over a long period of time

	Worker A	Worker B
Mean time for completion of job	36	25
Standard deviation	6	4

(a) Which worker has greater variability in completion of job
(b) Which worker is more consistent

Probability

- Probability is the study which deals with mathematical prediction of happening or non-happening of events.
- $\text{Probability} = \dfrac{\text{Number of events}}{\text{Number of sample}}$
- $\text{Probability} = \dfrac{n(E)}{n(S)}$

OBJECTS USED IN PROBABILITY

- Coin is the objects which contains two faces Head (H) and Tail (T)
- Die is the object which contains six (6) faces numbered 1,2,3,4,5 and 6.
- Tetrahedral is the object which contains four (4) faces numbered 1,2,,3, and 4.
- Cards are objects which occur together in the box of 52 playing cards and become 54 cards including jockers.

TERMS USED IN PROBABILITY

- **Random experiment** is one whose results depend on chance, that is the result cannot be predicted. Tossing of coins, throwing of dice are some examples of random experiments.
- **Trial** Performing a random experiment is called a trial.
- **Sample space** is a set of all possible outcomes in a random experiment.
- **Event:** An outcome or a combination of outcomes of a random experiment is called an event. For example tossing of a coin is a random experiment and getting a head or tail is an event
- **Outcomes:** The results of a random experiment are called its outcomes. When two coins are tossed the possible outcomes are HH, HT,TH,TT

TYPES OF EVENTS

- **Simple Event:** Simple Event is a single element of a sample space
- **Compound Event :**Compound Event is a joint occurrence of two or more events
- **Sure Event:**Sure Event is an event which has the same elements as that of the sample space
- **Impossible Event:**An Impossible Event is an event which has no possibility to occur
- **Mutually Exclusive Event:** is an event in which the occurrence of one event hinder or prevent the occurrence of the other events
- **Equally likely events:**Two or more events are said to be equally likely if each one of them has an equal chance of occurring. For example in tossing of a coin, the event of getting a head and the event of getting a tail are equally likely events.

TYPES OF EVENTS CONT...

- **Exhaustive events:** Events are said to be exhaustive when their totality includes all the possible outcomes of a random experiment. For example, while throwing a die, the possible outcomes are $\{1, 2, 3, 4, 5, 6\}$ and hence the number of cases is 6. or
- Exhaustive Events are events whose union forms the entire set (Sample space).
- Compliment of an Event is the set of all elements which are not in the event (E) but they are also found in the sample space.
- Equal Likely Events are events which have equal chances of occurrence.
- Independent Events are events in which the occurrence of one event does not affect the occurrence of the other event.
- **Dependent events:** If the occurrence of one event influences the occurrence of the other, then the second event is said to be dependent on the first

Fundamental of Counting

- The fundamental counting principle or counting rule is a way to figure out the number of outcomes in a probability problems
- Basically, you multiply the events together to get the total number of outcomes.
- If there are m ways to do one event and n ways to do another event, then there are mxn ways of doing both events

Fundamental principal of counting cont...

- "If one event can occur in e_1 ways, the second event can occur in e_2 the third event can occur in e_3 ways and the n^{th} event can occur in e_n ways, therefore the whole set or whole experiment can occur in $e_1 x e_2 x e_3 x ... x e_n$ ways"

Example

In how many ways can an organization containing 20 members elect a President, Treasurer and Secretary (assuming no person is elected more than one position)

Solution

President	Treasurer	secretary
20	19	18

Number of ways $= 20x19x18 = 6840$ways
number of ways 6840

- If a license plate contains three letters followed by 3 digits with the first digit non zero. How many different license plates can be printed?

Principle of permutation

- Permutation is the arrangement of the number of items taken from many items in which the order matters.
- Consider the letters A,B and C which are to be arranged in a group of two letters per time, how many different arrangement can be obtained
- AB,AC, BC, BA,CA,CB there are 6 arrangements.

- Let n be number of sample space (Total number of objects involved in experiment), and r be number of object taken per time.

Principle of permutaionn cont....

- by considering the above example three letters are arranged two letters per time how many different are arrangement are there? $n = 3, r = 2$
- Therefore the arrangement of $'r'$ objects taken from $'n'$ distinct objects is given $^np_r = \dfrac{n!}{(n-r)!}$

Note

- If there are n distinct objects and all n objects are arranged per time $n = r$
- The number of ways is given $^np_n = \dfrac{n!}{(n-n)!} = \dfrac{n!}{0!} = n!$, recall, $0! = 1$

- If there are n of which r_1 are of one kind, r_2 are of second kind, r_3 are of third kind,..., r_n are of the n^{th} kind,
- Therefore the total number ways is given by is given
$^np_r = \dfrac{n!}{r_1!xr_2!r_3x...r_n!}$

Arrangement of objects in a circular form

- The number of ways if $'n'$ unlike objects are arranged in a circular form when clockwise and anticlockwise arrangements are different is given $(n-1)!$
- Number of ways $= (n-1)!$
- The number of ways if $'n'$ unlike objects are arranged in a circular form when clockwise and anticlockwise arrangements are the same is given by $\dfrac{(n-1)}{2}!$
- Number of ways $= \dfrac{(n-1)!}{2}$

Example

- In how many ways can a manager display 5 brands of cereals in 3 spaces on a shelf?
  Solution
  Given $n = 5$ and $r = 3$
  $^{n}p_r = \dfrac{n!}{(n-r)!} = \dfrac{5!}{(5-3)!} = 60$
- In how many ways can 5 people be arranged in a row?
- In how many ways can 4 people be arranged in a circle?
- How many different ways can 4 red,3 yellow and 2 blue bulbs be arranged in a string of Christmas tree lights with 9 sockets?
- How many 4 digit numbers can we make using the digits 3 ,6 ,7 and 8 without repetitions?

Principle of combination

- Combination is the selection of the number of items taken from many items in which the order does not matter.
- Consider a group of two letters selected from the group of three letters A,B and C then the selection (number of combination) are AB, AC and BC, there are 3 selections or group
- The number ways of choosing a set of $'r'$ objects from $'n'$ distinct objects is given by ${}^{n}C_{r} = \frac{{}^{n}p_{r}}{r!}$
- ${}^{n}C_{r} = \frac{n!}{(n-r)!r!}$

example

- In how many ways can a coach choose three swimmers from among five swimmers?
  Solution
  Given $n=5$ and $r=3$
  $^nC_r = \dfrac{n!}{(n-r)!r!} = dfrac5!(5-3!3! = 10$
  A coach can choose the swimmers in 10 ways

example

- How many triangles can you make using 6 non collinear points on a plane?
- Suppose we have 12 adults and 10 kids as an audience of a certain show. Find the number of ways the host can select three persons from the audiences to volunteer. The choice must contain two kids and one adult.
- From a group of 7 men and 6 women, 5 persons are to be selected to form a committee so that at least 3 men are there in the committee. In how many ways can it be done?
- Out of 10 mathematicians, a committee of 5 mathematicians has to be made. In how many ways can this be done?

PROBABILITY AXIOMS AND THEOREM

- If A and B are two events associated with an experiment.
- Then, $P(A\cup B)=P(A)+P(B)-P(A\cap B)$
  Proof
  consider $n(A\cup B)=n(A)+(B)-(A\cap B)$
  Divide by sample space $n(S)$ both sides
  $$\frac{n(A\cup B)}{n(S)}=\frac{n(A)}{n(S)}+\frac{n(B)}{n(S)}-\frac{n(A\cup B)}{n(S)}$$
  $P(A\cup B)=P(A)+P(B)-P(A\cap B)$
- If A and B are two disjoint events $P(A\cup B)=P(A)+P(B)$
- Disjoint events are events which does not intersect i.e.$P(A\cap B)=0$
- If A and B are two independent events, then $P(A\cup B)=P(A)xP(B)$

- If E is an event and E' is compliment of an event, then
  $P(E)+P(E')=1$
  Proof
  Consider $n(E)+n(E')=n(S)$
  Divide by sample space $n(S)$ both sides
  $$\frac{n(E)}{n(S)}+\frac{n(E')}{n(S)}=1$$
  likewise
- $n(A\cup B)+n(A\cup B)^{'}=1$
  and
- $n(A\cap B)+n(A\cap B)^{'}=1$

- $P(A \cup B)' = P(A' \cap B')$
  Proof
  Consider $n(A \cap B)' = n(A' \cup B')$
  Divide by sample space $n(S)$ both sides
  $\frac{n(A \cap B)'}{n(S)} = \frac{n(A' \cup B')}{n(S)}$
  likewise
- $P(A \cup B)' = P(A' \cap B')$
  and
- $P(A \cap B)' = P(A' \cup B')$

- $P(A) = P(A \cap B) + P(A \cap B')$

  Proof

  Consider $n(A) = n(A \cap B) + (A \cap B')$

  Divide by sample space $n(S)$ both sides

  $\frac{n(A}{n(S)} = \frac{n(A \cap B)}{n(S)} + \frac{(A \cap B')}{n(S)}$

  $P(A) = P(A \cap B) + P(A \cap B')$ likewise

- $P(B) = P(A \cap B) + P(A' \cap B)$

- $P(A') = P(A' \cap B') + P(A' \cap B)$
  Proof
  Consider $n(A) = n(A' \cap B') + (A' \cap B)$
  Divide by sample space $n(S)$ both sides
  $\frac{n(A'}{n(S)} = \frac{n(A' \cap B')}{n(S)} + \frac{(A' \cap B)}{n(S)}$
  $P(A') = P(A' \cap B') + P(A' \cap B)$ likewise
- $P(B') = P(A' \cap B') + P(A \cap B')$

Summary

	B	B'	
A	$P(A \cap B)$	$P(A \cap B')$	$P(A)$
A'	$P(A' \cap B)$	$P(A' \cap B')$	$P(A')$
	$P(B)$	$P(B')$	$P(S) = 1.0$

If $P(A) = 0.5, P(B) = 0.3$ and $P(A \cap B) = 0.2$ Find

(i). $P(A \cap B')$

(ii). $P(A' \cap B)$

(iii). $P(A \cup B)$

(iv). $P(A' \cap B')$

Solution

(i). $P(A \cap B') = P(A) - P(A \cap B) = 0.5 - 0.2 = 0.3$

(ii). $P(A' \cap B) = P(B) - P(A \cap B) = 0.3 - 0.2 = 0.1$

(iii). $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.5 + 0.3 - 0.2 = 0.6$

(iv). $P(A' \cap B') = 1 - P(A \cup B) = 1 - 0.6 = 0.4$

Addition theorem on probabilities for Mutually Exclusive Events:

- If two events A and B are mutually exclusive the probability of the occurrence of either A or B is the sum of the individual probability of A and B $P(A \cup B) = P(A) + P(B)$

Addition theorem on probabilities not mutually exclusive events:

- If two events A and B are not mutually exclusive the probability of the occurrence of either A or B is the sum of the individual probability of minus probability for both to happen $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Example

If $P(A) = 0.5, P(B) = 0.6$ and $P(A \cap B) = 0.2$ Find

(i). $P(A \cup B)$

(ii). $P(A')$

(iii). $P(A \cap B')$

(iv). $P(A' \cap B')$

Multiplication theorem on probabilities for independent events:

- If two events A and B are independent, the probability that both of them occur is equal to the product of their individual probabilities$P(A \cap B) = P(A)xP(B)$

Multiplication theorem on probabilities for dependent events:

- If two events A and B are dependent, the probability of occurring 2^{nd} event will be affected by the outcome of the first event. $P(A \cap B) = P(A)xP(B/A)$

Example

A bag contains 5 white balls and 8 black balls. one ball is drawn at random from the bag. Again , another one is drawn without replacing the first ball. Find the probability that both balls drawn are white.

CONDITIONAL PROBABILITY

- Conditional probability is the probability in which occurrence of one event depends on the occurrence of another event.
- Conditional probability is the probability of an event given some other event has already occurred.
- Consider the probability of an event B given that an event A has already occurred.
- The probability of an event in relation to the other event is called conditional probability of B given$A$.
- Conditional probability of B given A is denoted by $P(B/A)$ that is A occurs before B.

CONDITIONAL PROBABILITY CONT...

- In a case where events$A$ and$B$ are independent (where event A has no effect on the probability of event$B$), the conditional probability of event B given event A is simply the probability of event B, that is $P(B)$
- If events A and B are dependent, the probability of the intersection of A and B (the probability of both events occurs) is defined by$P(A \cap B) = P(A)P(B/A)$
- from this definition the conditional probability is easily obtained by dividing by $P(A)$
- The conditional probability of an event Bgiven that A has already taken place is defined mathematically as$P(B/A) = \dfrac{p(A \cap B)}{P(A)}$ Note: $P(A) \neq 0$

Example

- In a class, 40% of the students study math and science. 60% of the students study math. What is the probability of a student studying science given he/she is already studying math?
  Solution
  Let $A =$ Students who study Math
  $B =$ Students who study Science
  $P(A \cap B) = 40\% = 0.4$
  $P(A) = 60\% = 0.6$
  Required$P(B/A)$?
  $$P(B/A) = \frac{p(A \cap B)}{P(A)} = \frac{0.4}{0.6} = \frac{2}{3}$$

example

In an exam, two reasoning problems, 1 and 2, are asked. 35% students solved problem 1 and 15% students solved both the problems. How many students who solved the first problem will also solve the second one?

example

Out of 50 people surveyed in a study, 35 smoke in which there are 20 males. What is the probability that if the person surveyed is a smoker then he is a male?

BAYES' Theorem:

- The concept of conditional probability discussed earlier takes into account information about the occurrence of one event to predict the probability of another event.
- This concept can be extended to revise probabilities based on new information and to determine the probability that a particular effect was due to specific cause.
- The procedure for revising these probabilities is known as Bayes theorem.

Bayes' Theorem or Rule (Statement only):

- Let $A_1, A_2, A_3, \ldots\ldots\ldots A_i, \ldots\ldots A_n$ be a set of n mutually exclusive and collectively exhaustive events and$P(A_1), P(A_2)\ldots, P(A_n)$ are their corresponding probabilities. If B is another event such that P(B) is not zero and the priori probabilities $P(B|A_i) i = 1, 2\ldots, n$ are also known. Then $P(A_i/B) = \dfrac{P(B/A_i)P(A_i)}{\sum_{i=1}^{k} P(B/A_i)P(A_i)}$

Random Variables

- A random variable is a variable whose value is a numerical outcome of a random phenomenon usually denoted by capital letters X or Y.

Types of Random Variables

- **Discrete Random Variables**
- **Continuous Random Variables**

Discrete Random Variables

- A random variable X is said to be discrete if it can assume only a finite or countably infinite number of distinct values. e.g. Bernoulli, Binomial, Geometric, Hypergeometric, Negative binomial, and Poisson random variable.
- Discrete Random Variable is random variable which takes countable or finite number of values eg 1.2.3....
- Discrete random variable may take on only a countable number of distinct values

- The probability distribution of a discrete random variable is a list of probabilities associated with each of its possible values.
- It is also sometimes called the probability function or the probability mass function. Let X be a random variable which can take at most a finite number of values$x_1, x_2, x_3, ...$
- If the probability that X takes the values x_i is $P(x_i)$ then

$$\begin{cases} P(x_i) \geq 0 \\ \sum P(x_i) = 1 \end{cases}$$

- All random variables (discrete and continuous) have a cumulative distribution function.
- It is a function giving the probability that the random variable X is less than or equal to x .
- For a discrete random variable, the cumulative distribution function is found by summing up the probabilities.

Mathematical Expectation

- The expectation or expected value of the random variable is the mean (average) of the probability distribution.
- If X is a discrete random variable which can take values $x_1, x_2, x_3, ...x_n$ with their probabilities $P(x_1), P(x_2), P(x_3)...P(x_n)$ respectively
- Then the mathematical expectation is denoted by $E(x)$ is given $E(x) = x_1 P(x_1) + x_2 P(x_2) + x_3 P(x_3) + ... + x_n P(x_n) = \sum_{i=1}^{n} x_i P(x_i)$

Properties of Expectation

(i). $E(a) = a$

(ii). $E(ax) = aE(a)$

(iii). $E(ax + b) = aE(x) + b$

(iv). $E(g(x) + h(x)) = E(g(x)) + E(h(x))$

(v). $E[g(x).h(x)] = E(g(x).E(h(x))$

Variance of Discrete random variable

- In probability and statistics, the variance of a random variable is the average value of the square distance from the mean value.
- It represents the how the random variable is distributed near the mean value. Small variance indicates that the random variable is distributed near the mean value.
- Big variance indicates that the random variable is distributed far from the mean value.
- The variance of random variable X is the expected value of squares of difference of X and the expected value μ
- The variance of X denoted by $Var(X)$ and is defined as $Var(X) = E(X-\mu)^2$ where X is the random variable and μ is the mean.

Properties of Variance

(i). $Var(a) = 0$(a is a constant and never vary)

(ii). $Var(ax) = a^2E(x)$

(iii). $Var(ax + b) = a^2Var(x)$

(iv). $Var[g(x) + h(x)] = Var[g(x)] + Var[h(x)]$

Example

The p.d.f of a discrete random variable X is given by $P(X = x) = cx^2$
For $0, 1, 2, 3$. find the value of constant c

Solution

$X = x$	0	1	2	3
$P(X = x)$	0	c	4c	9c

Example

random variable X has a p.d.f for as shown below;

$X = x$	1	2	3
$P(X = x)$	0.1	0.6	0.3

Find

(a). $E(x)$

(b). $E(5)$

(c). $E(4x)$

(d). $E(3x + 2)$

(e). $E(x^2)$

(g). $E(2x^2)$

(h). $E(2x^2) + 3$

Example

A discrete random variable X has the following probability distribution:

x	10	15	20	25
$p(x)$	0.2	0.3	0.4	0.1

Calculate the variance and standard deviation.
Solution

$$E[X] = \sum_x xp(x)$$
$$= 10(0.2) + 15(03) + 20(0.4) + 25(0.1) = 17$$
$$E[X] = 17$$
$$\sigma^2 = \text{Var}(X) = \sum_x x^2 p(x) - \mu^2$$
$$= 10^2(0.2) + 15^2(03) + 20^2(0.4) + 25^2(0.1) - 17^2 = 310 - 289 = 21$$
$$\sigma = 4.5826$$

Example

A discrete random variable X has a probability function given by

$$P(x) = \begin{cases} kx^2 & x = 1, 2, 3. \\ 0 & otherwise \end{cases}$$

Find

(a). The value of k

(b). $E(x^2)$

(c). Var(x)

(d). S.d of x

Continuous Random Variables

- A random variable X is said to be continuous if it assume uncountable set of possible values. e.g. Uniform, Exponential, Gamma, Beta, Chi-square and Normal random variable.
- Continuous Random Variable is a random variable which takes an infinite number of possible values.
- Continuous random variables are usually measurements. Examples includes height, weight, the amount of sugar in an orange, the time required to run a mile.

Continuous Probability Distributions

- The probability distribution (probability density function, pdf) of X on the interval (a, b) is given by

$$P(a \leq X \leq b) = \int_a^b f(x)\mathrm{d}x.$$

- There are two conditions for a given function f to be a p.d.f
  1. $f(x) \geq 0$ for all values of x.
  2. $\int_{-\infty}^{\infty} f(x)\mathrm{d}x = 1$
- The **cumulative distribution function (cdf)** is given by

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\mathrm{d}t.$$

- If f is the pdf of a random variable X on the interval [a, b], then

$$P(a \leq X \leq b) = \int_a^b f(x)\mathrm{d}x.$$

continuous Probability Distributions cont...

- Some properties of cumulative distribution function F
  1. $0 \leq F(x) \leq 1$.
  2. $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to \infty} F(x) = 1$.
  3. F is a nondecreasing function, and right continuous.

Expectation of contentious variable

The expectation of a continuous random variable X is defined as $\mu = E[X] = \int_{-\infty}^{\infty} xf(x)\mathrm{d}x$.

Properties of Expectation

(i). $E(a) = a$

(ii). $E(ax) = aE(a)$

(iii). $E(ax + b) = aE(x) + b$

(iv). $E(g(x) + h(x)) = E(g(x)) + E(h(x))$

(v). $E[g(x).h(x)] = E(g(x).E(h(x))$

Variance of continuous random variable

- In probability and statistics, the variance of a random variable is the average value of the square distance from the mean value.
- It represents the how the random variable is distributed near the mean value. Small variance indicates that the random variable is distributed near the mean value.
- Big variance indicates that the random variable is distributed far from the mean value.
- The variance of random variable X is the expected value of squares of difference of X and the expected value μ
- The variance of X denoted by $Var(X)$ and is defined as $Var(X) = E(X-\mu)^2$ where X is the random variable and μ is the mean.

- $Var(X) = E(x-\mu)^2$
- $Var(X) = E(x^2) - (E(x))^2$
- $Var(X) = \int x^2 f(x)dx - (\int xf(x)dx)^2$

Properties of Variance

(i). $Var(a) = 0$(a is a constant and never vary)

(ii). $Var(ax) = a^2 E(x)$

(iii). $Var(ax+b) = a^2 Var(x)$

(iv). $Var[g(x)+h(x)] = Var[g(x)] + Var[h(x)]$

Example

- A continuous random variable X has a p.d.f $f(x) = kx^2$ for $0 \leq x \leq 4$

(a). Find the value of k

(b). find $P(1 \leq x \leq 3)$

Example

- X is a delay in hours of a flight from Chicago where $f(x) = 0.2 - 0.02x$ for $0 \leq x \leq 10$ Find,

(a). The probability that the delay be less than 4 hours

(b). The probability of the delay be between 2 and 6 hours

Example

- Let the function $f(x) = \begin{cases} cx^2 & \text{if} \quad 0 < x < 3 \\ 0 & \text{otherwise/elsewhere} \end{cases}$

  i. Find the value of c so that $f(x)$ is a density function.f?

  ii. Compute $P(2 < X < 3)$.

Solution

i. Formula $\int_{-\infty}^{\infty} f(x)\mathrm{d}x = 1$

$$\int_0^3 cx^2 \mathrm{d}x = 1,$$

$$\left.\frac{cx^3}{3}\right|_0^3 = 1,$$

$$9c = 1$$

$\therefore$ The value of $c = \frac{1}{9}$.

ii. The $P(2 < X < 3)$ is the probability between 2 and 3 inclusive

$$
\begin{aligned}
P(2 < X < 3) &= \int_2^3 \frac{x^2}{9}\,\mathrm{d}x \\
&= \left.\frac{x^3}{27}\right|_2^3 \\
&= \frac{27}{27} - \frac{8}{27} \\
\therefore P(2 < X < 3) &= \frac{19}{27}
\end{aligned}
$$

- The amount of time, in hours, that a machine functions before breakdown is a continuous random variable with pdf

$$f(t) = \begin{cases} \frac{1}{120} e^{-t/120}, & t \geq 0 \\ 0, & t < 0. \end{cases}$$

  i. What is the probability that it will function less than 160 hours?
  ii. What is the probability that this machine will function between 98 and 145 hours before breaking down?

Solution

i. $P(T < 160)$

$$P(0 < T < 160) = \int_0^{160} \frac{1}{120} e^{-t/120} \mathrm{d}t$$

$$\text{Let } u = \frac{-t}{120}, t = 0, \Rightarrow u = 0, t = 160, \Rightarrow u = \frac{-4}{3}$$

$$\mathrm{d}u = \frac{-\mathrm{d}t}{120} \Rightarrow \mathrm{d}t = -120\mathrm{d}u,$$

$$\begin{aligned}
\int_0^{160} \frac{1}{120} e^{-t/120} \mathrm{d}t &= \int_0^{-4/3} \frac{1}{120} e^{u} (-120\mathrm{d}u), \\
&= \int_0^{-4/3} -e^{u} \mathrm{d}u, \\
&= -e^{u}\Big|_0^{-4/3} = -[e^{-4/3} - 1], \\
P(0 < T < 160)e &= 1 - e^{-4/3} = 1 - 0.2636 = 0.7364.
\end{aligned}$$

Thus the probability that it will function less than 160 hours is 0.7364.

ii.$P(98 < T < 145)$

$$\begin{aligned} P(98 < T < 145) &= \int_{98}^{145} \frac{1}{120} e^{-t/120} \mathrm{d}t \\ &= -e^{-t/120}\Big|_{98}^{145} \\ &= -(e^{-145/120} - e^{-98/120}) = 0.1432. \end{aligned}$$

Therefore the probability that this machine will function between 98 and 145 hours before breaking down is 0.1432.

Example

X is a continuous variable, the mass in kg of a substance produced per minute in industrial process where;

$$f(x) = \begin{cases} \frac{1}{36}x(6-x) & 0 \le x \le 6 \\ 0 & otherwise \end{cases}$$

Find the probability that the mass is more than 5kg.

Example

- The continuous random variable X has a p.d.f $f(x)$ where $f(x) = \frac{1}{8}x$ for $0 \le x \le 4$ Find,

(a). $E(x)$

(b). $Var(x)$ and $Var(3x)$

(c). Standard deviation

(d). $Var(3x+10)$

Example

An exponential random variable with parameter λ is given by the *pdf*

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases} \qquad \text{Find } Var(X).$$

Solution

$$\begin{aligned} \text{Var}(X) &= \int_{-\infty}^{\infty} x^2 f(x)\mathrm{d}x - \mu^2 \\ &= \int_0^{\infty} \lambda x^2 e^{-\lambda x}\mathrm{d}x - \frac{1}{\lambda^2}, \quad \text{using integration by parts twice yieds,} \end{aligned}$$

$$\text{Var}(X) = \frac{1}{\lambda^2}.$$

Questions

1. If X is uniform on the interval [a, b] then find its variance.
2. An exponential random variable with parameter β is given by the *pdf*

$$f(x) = \begin{cases} \frac{1}{\beta} e^{-x/\beta}, & x \geq 0 \\ 0, & x < 0. \end{cases} \qquad \text{Find Var}(X).$$

Moment-generating function of a random variable

- For a random variable X, suppose that there is a positive number h such that for $-h < t < h$ the mathematical expectation $E[e^{tX}]$ exists.
- The **moment-generating function (mgf)** of the random variable X is defined by

$$M_X(t) = E[e^{tX}] = \sum e^{tX} p(x), \qquad \text{if X is discrete and}$$
$$M_X(t) = E[e^{tX}] = \int e^{tX} f(x)\mathrm{d}x \qquad \text{if X is continuous}$$

- We call $M_X(t)$ the moment generating function because all of the moments of X can be obtained by successively differentiating $M_X(t)$.

- For example,

$$
\begin{aligned}
M_X'(t) &= \frac{d}{dt}E[e^{tX}] \\
&= E\frac{d}{dt}[e^{tX}] \\
&= E[Xe^{tX}] \quad \text{put } t = 0 \\
M_X'(0) &= E[X]
\end{aligned}
$$

Similarly

$$
\begin{aligned}
M_X''(t) &= \frac{d}{dt}M_X'(t) \\
&= E\frac{d}{dt}[Xe^{tX}] \\
&= E[X^2 e^{tX}] \quad \text{put } t = 0 \\
M_X''(0) &= E[X^2]
\end{aligned}
$$

- In general, the nth derivative of $M_X(t)$ evaluated at $t = 0$ equals

- In general, the n^{th} derivative of $M_X(t)$ evaluated at t = 0 equals $E[X^n]$, that is
$$M_X^n(0) = E[X^n],\ n \geq 1$$

Example

- If X is uniform on the interval [a, b] then show that the mean, variance and moment generating function of X are given by

$$E[X] = \frac{b+a}{2}$$

$$\text{Var}(X) = \frac{1}{12}(b-a)^2$$

$$M(t) = \begin{cases} \frac{e^{tb}-e^{ta}}{t(b-a)} & \text{if } t \neq 0 \\ 1 & \text{if } t = 0 \end{cases}$$

- Let X be a random variable with pdf given by
  $f(x) = \begin{cases} \frac{1}{\beta} e^{-x/\beta}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$
  Find mgf $M_X(t)$.

Joint Probability Distributions

- **Discrete case**
- The joint probability function of two discrete random variables X and Y is defined by
  $P(X=x, Y=y) = f(x,y)$ where
  1. $f(x,y) \geq 0$
  2. $\sum_x \sum_y (fx,y) = 1$. i.e., the sum over all values of x and y is 1.
- **Continuous case**
  The joint probability function for two continuous random variables X and Y (or, as it is more commonly called, the joint density function of X and Y) is defined by
  1. $f(x,y) \geq 0$
  2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) \mathrm{d}y \mathrm{d}x = 1$
- The probability under the interval $a < X < b, c < Y < d$ is given by,

$$P(a < X < b, c < Y < d) = \int_a^b \int_c^d f(x,y) \mathrm{d}y \mathrm{d}x.$$

Example

- Suppose that the random variables X and Y have a joint density function given by

$$f(x,y) = \begin{cases} c(2x+y), & 2 < x < 6, 0 < y < 5, \\ 0, & \text{otherwise} \end{cases}$$

(a) Find the constant c.
(b) Find $P(3 < X < 4, Y > 2)$.

Solution

(a) The total probability is given by

$$\int_{x=2}^{6}\int_{y=0}^{5} c(2x+y)\mathrm{d}y\mathrm{d}x = 1,$$

$$\int_{x=2}^{6} c(2xy+\frac{y^2}{2})\Big|_{y=0}^{5}\mathrm{d}x = 1,$$

$$\int_{x=2}^{6} c(10x+\frac{25}{2})\mathrm{d}x = 1,$$

$$c(5x^2+\frac{25x}{2})\Big|_{2}^{6} = 1,$$

$$c\,[(180+75)-(100+25)] = 1$$

$$210c = 1$$

Thus the value of $c = \frac{1}{210}$.

(b) The total probability is given by

$$
\begin{aligned}
P(3 < X < 4, Y > 2) &= \frac{1}{210}\int_{x=3}^{4}\int_{y=2}^{5}(2x+y)\mathrm{d}y\mathrm{d}x,\\
&= \frac{1}{210}\int_{x=3}^{4}(2xy+\frac{y^2}{2})\Big|_{y=2}^{5}\mathrm{d}x,\\
&= \frac{1}{210}\int_{x=3}^{4}(6x+\frac{21}{2})\mathrm{d}x,\\
&= \frac{1}{210}(3x^2+\frac{21x}{2})\Big]_{x=3}^{4},\\
P(3 < X < 4, Y > 2) &= \frac{3}{20.}
\end{aligned}
$$

Line and Multiple Integrals

- **Evaluation of Line Integrals**
- **Method of Evaluation-Curve Defined Parametrically**
  Let $z = G(x, y)$ be defined in some region that contains the smooth curve C, If C is a smooth curve parameterised by $x = f(t), y = g(t), a \leq t \leq b$, then we simply replace x and y in the integral by the functions $f(t)$ and $g(t)$, and the appropriate differential dx, dy, or ds by $f'(t)dt, g'(t)dt$, or $\sqrt{[f'(t)]^2 + [g'(t)]^2}\mathrm{d}t$
- The expression $ds = \sqrt{[f'(t)]^2 + [g'(t)]^2}\mathrm{d}t$ is called the differential of arc length.

$$\int_C G(x,y)\mathrm{d}x = \int_a^b G(f(t), g(t))\, f'(t)\mathrm{d}t$$
$$\int_C G(x,y)\mathrm{d}y = \int_a^b G(f(t), g(t))\, g'(t)\mathrm{d}t$$
$$\int_C G(x,y)\mathrm{d}s = \int_a^b G(f(t), g(t))\sqrt{[f'(t)]^2 + [g'(t)]^2}\mathrm{d}t.$$

Example

- Evaluate (a) $\int_C xy^2\mathrm{d}x$ (b) $\int_C xy^2\mathrm{d}y$ and $\int_C xy^2\mathrm{d}s$ on the quarter-circle C defined by $x = 4\cos\,t,\; y = 4\sin\,t, 0 \le t \le \pi/2$.
- **Solution**

$$(a) \int_C G(x,y)\mathrm{d}x = \int_0^{\pi/2} (4\cos\,t)\left(16\sin^2 t\right)(-4\sin\,t\mathrm{d}t),$$

$$= -256\int_0^{\pi/2} \sin^3 t\cos\,t\mathrm{d}t,$$

$$= -256.\,\frac{1}{4}\sin^4 t\mathrm{d}t\Big|_0^{\pi/2},$$

$$\therefore \int_C G(x,y)\mathrm{d}x = -64.$$

$$
\begin{aligned}
\text{(b)} \int_C G(x,y)\mathrm{d}y &= \int_0^{\pi/2} (4\cos\, t)\left(16\sin^2 t\right)(4\cos\, t\mathrm{d}t),\\
&= 256\int_0^{\pi/2} \sin^2 t\cos^2 t\mathrm{d}t = 256\int_0^{\pi/2} \frac{1}{2}(2\sin\, t\cos\, t).\frac{1}{2}(\\
&= 256\int_0^{\pi/2} \frac{1}{4}\sin 2t.\sin 2t\mathrm{d}t = 256\int_0^{\pi/2} \frac{1}{4}\sin^2 2t\mathrm{d}t,\\
&= 64\int_0^{\pi/2} \frac{1}{2}(1-\cos 4t)\,\mathrm{d}t,\\
&= 32\left[t-\frac{1}{4}\sin 4t\right]\Bigg|_0^{\pi/2}\\
\therefore \int_C G(x,y)\mathrm{d}y &= 16\pi.
\end{aligned}
$$

$$\text{(c)} \int_C G(x,y)\mathrm{d}s = \int_0^{\pi/2} (4\cos\, t)\left(16\sin^2 t\right)\sqrt{16\left(\cos^2 t + \sin^2 t\right)}\mathrm{d}t,$$

$$= 256\int_0^{\pi/2} \sin^2 t\cos\, t\mathrm{d}t,$$

$$= 256.\, \frac{1}{3}\sin^3 t\mathrm{d}t\Big|_0^{\pi/2},$$

$$\therefore \int_C G(x,y)\mathrm{d}s = \frac{256}{3}.$$

Method of Evaluation-Curve Defined by an Explicit Function

- If the curve C is defined by an explicit function $y = f(x), a \leq x \leq b$, we can use x as a parameter. With $dy = f'(x)dx$ and ds $=\sqrt{1+[f'(x)]^2}dx$, the foregoing line integrals become, in turn,
- $$\int_C G(x,y)\mathrm{d}x = \int_a^b G(x, f(x))\,\mathrm{d}x$$
  $$\int_C G(x,y)\mathrm{d}y = \int_a^b G(x, f(x))\,f'(x)\mathrm{d}x$$
  $$\int_C G(x,y)\mathrm{d}s = \int_a^b G(x, f(x))\,\sqrt{[1+[f'(x)]^2}\mathrm{d}x.$$

Example

- Evaluate $\int_C xy\mathrm{d}x + x^2\mathrm{d}y$ where C is given by $y = x^3$, $-1 \leq x \leq 2$.
- **Solution**
-

$$\begin{aligned}
y = x^3 \quad &\Rightarrow \mathrm{d}y = 3x^2\mathrm{d}x \\
\int_C xy\mathrm{d}x + x^2\mathrm{d}y &= \int_{-1}^{2} x(x^3)\mathrm{d}x + x^2(3x^2)\mathrm{d}x, \\
&= \int_{-1}^{2} 4x^4\mathrm{d}x, \\
&= \frac{4}{5}x^5\Big|_{-1}^{2}, \\
\therefore \int_C xy\mathrm{d}x + x^2\mathrm{d}y &= \frac{132}{5}.
\end{aligned}$$

Example

- Evaluate the line integral $\int_C -4x\mathrm{d}x + y^2\mathrm{d}y - yz\mathrm{d}z$ with C given by $x = -t^2, y = 0, z = -3t$ for $0 \le t \le 1$.
- **Solution**
-

$$x = -t^2, \quad dx = -2tdt,$$
$$y = 0, \quad dy = 0,$$
$$z = -3t, \quad dz = -3dt.$$

$$\begin{aligned}\int_C -4x\mathrm{d}x + y^2\mathrm{d}y - yz\mathrm{d}z &= \int_0^1 -4(-t^2)(-2t\mathrm{d}t) + 0 - 0,\\ &= \int_0^1 -8t^3\mathrm{d}t = -8\int_0^1 t^3\mathrm{d}t,\\ &= -8 \cdot \frac{t^4}{4}\bigg|_0^1,\end{aligned}$$

$$\int_C -4x\mathrm{d}x + y^2\mathrm{d}y - yz\mathrm{d}z = -2.$$

Line Integral in Vector Notation

- Line integral in 2-Space
  Let $\mathbf{F}(x,y) = P(x,y)\mathbf{i} + Q(x,y)\mathbf{j}$ is a **vector field/force field** in 2-space and a path C is defined by the vector function $\mathbf{r}(t) = x\mathbf{i} + y\mathbf{j}, x = x(t), y = y(t), a \leq t \leq b$. Thus
- $$\mathrm{d}\mathbf{r} = \mathrm{d}x\mathbf{i} + \mathrm{d}y\mathbf{j}, \mathrm{d}x = x'(t)\mathrm{d}t, \mathrm{d}y = y'(t)\mathrm{d}t,$$
  $$\mathbf{F}.\mathrm{d}\mathbf{r} = (P(x,y)\mathbf{i} + Q(x,y)\mathbf{j}).(\mathrm{d}x\mathbf{i} + \mathrm{d}y\mathbf{j}) = P(x,y)\mathrm{d}x + Q(x,y)\mathrm{d}y$$
- then a line integral is defined as
  $$\int_C \mathbf{F}.\mathrm{d}\mathbf{r} = \int_C P(x,y)\mathrm{d}x + Q(x,y)\mathrm{d}y$$

Line integral in 3-Space

- Let $\mathbf{F}(x,y,z) = P(x,y,z)\mathbf{i} + Q(x,y,z)\mathbf{j} + R(x,y,z)$ is a **vector field/force field** in 3-space and a path C is defined by the vector function $\mathbf{r}(t) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}, x = x(t), y = y(t), z = z(t), a \leq t \leq b$. Thus
-

$$\begin{aligned}
\mathrm{d}\mathbf{r} &= \mathrm{d}x\mathbf{i} + \mathrm{d}y\mathbf{j} + \mathrm{d}z\mathbf{k}, \mathrm{d}x = x'(t)\mathrm{d}t, \mathrm{d}y = y'(t)\mathrm{d}t, \mathrm{d}z = z'(t)dt,\\
\mathbf{F}.\mathrm{d}\mathbf{r} &= (P(x,y,z)\mathbf{i} + Q(x,y,z)\mathbf{j} + R(x,y,z)\mathbf{k}).(\mathrm{d}x\mathbf{i} + \mathrm{d}y\mathbf{j} + \mathrm{d}z\mathbf{k}),\\
&= P(x,y,z)\mathrm{d}x + Q(x,y,z)\mathrm{d}y + R(x,y,z)\mathrm{d}z
\end{aligned}$$

- then a line integral is defined as

$$\int_C \mathbf{F}.\mathrm{d}\mathbf{r} = \int_C P(x,y,z)\mathrm{d}x + Q(x,y,z)\mathrm{d}y + R(x,y,z)\mathrm{d}z.$$

In general, the **work done W** by a **force F along a curve C** is given by

$$W = \int_C \mathbf{F}.\mathrm{d}\mathbf{r}$$

Examples

- 1. Find the work done by a force $\mathbf{F}(x,y,z) = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$ acting along the curve given by $\mathbf{r}(t) = t^3\mathbf{i} + t^2\mathbf{j} + t\mathbf{k}$ from $t = 1$ to $t = 3$.
- **Solution**

$$\begin{aligned}
x &= t^3, \qquad y = t^2, \qquad z = t. \\
\mathrm{d}x &= 3t^2\mathrm{d}t, \quad \mathrm{d}y = 2t\mathrm{d}t, \; \mathrm{d}z = \mathrm{d}t. \\
W &= \int_C \mathbf{F}.\mathrm{d}\mathbf{r} = \int_C yz\mathrm{d}x + xz\mathrm{d}y + xy\mathrm{d}z, \\
&= \int_1^3 t^2.t.3t^2\mathrm{d}t + t^3.t.2t\mathrm{d}t + t^3.t^2.\mathrm{d}t, \\
&= \int_1^3 (3t^5 + 2t^5 + t^5)\mathrm{d}t = \int_1^3 6t^5\mathrm{d}t, \\
W &= t^6\Big]_1^3 = 3^6 - 1^6 = 728.
\end{aligned}$$

- $\therefore$ The work done is 728.

- 2. Evaluate the line integral $\int_C \mathbf{F}.\mathrm{d}\mathbf{R}$ with $\mathbf{F} = x\mathbf{i} + y\mathbf{j} - z\mathbf{k}$ and and C the circle $x^2 + y^2 = 4, z = 0$, going around once counterclockwise.
  Solution
  Parametrise the curve C, i.e the circle is parametrised as
  $\mathbf{R} = 2\cos t\mathbf{i} + 2\sin t\mathbf{j} + 0\mathbf{k}$,

$$
\begin{aligned}
x &= 2\cos t, y = 2\sin t, z = 0, 0 \leq t \leq 2\pi. \\
\mathrm{d}x &= -2\sin t\mathrm{d}t, dy = 2\cos t\mathrm{d}t, \mathrm{d}z = 0. \\
\int_C \mathbf{F}.d\mathbf{R} &= \int_C x\mathrm{d}x + y\mathrm{d}y - z\mathrm{d}z, \\
&= \int_0^{2\pi} 2\cos t(-2\sin t\mathrm{d}t) + 2\sin t(2\cos t\mathrm{d}t) + 0, \\
&= \int_0^{2\pi} -4\cos t\sin t\mathrm{d}t + 4\sin t\cos t\mathrm{d}t = 0, \\
\therefore \int_C \mathbf{F}.d\mathbf{R} &= 0.
\end{aligned}
$$

Vector Form of Green's Theorem

- Let $\mathbf{F}(x,y) = P(x,y)\mathbf{i} + Q(x,y)\mathbf{j}$ is a **vector field/force field** in 2-space and a path C is defined by the vector function $\mathbf{r}(t) = x\mathbf{i} + y\mathbf{j}, x = x(t), y = y(t), a \leq t \leq b$. If **C encloses region D** then the Green's Theorem

$$\oint_C \mathbf{F}.\mathrm{d}\mathbf{r} = \oint_C P\mathrm{d}x + Q\mathrm{d}y = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathrm{d}A.$$

- Conclusion

$$\oint_C P\mathrm{d}x + Q\mathrm{d}y = \begin{cases} 0, & \text{if C does not enclose the origin.} \\ 2\pi, & \text{if C encloses the origin.} \end{cases}$$

Example

- Use Green's theorem to evaluate $\oint_C \mathbf{F}.\mathrm{d}\mathbf{R}$, $\mathbf{F} = e^x\cos(y)\mathbf{i} - e^x\sin(y)\mathbf{j}$ and C is any simple closed positively oriented path in the plane.
- **Solution**
-

$$\oint_C \mathbf{F}.\mathrm{d}\mathbf{R} = \oint_C e^x\cos(y)\mathrm{d}x - e^x\sin(y)\mathrm{d}y,$$

$$P = e^x\cos(y) \quad \Rightarrow \frac{\partial P}{\partial y} = -e^x\sin(y),$$

$$Q = e^x\sin(y) \quad \Rightarrow \frac{\partial Q}{\partial x} = -e^x\sin(y),$$

$$\oint_C \mathbf{F}.\mathrm{d}\mathbf{R} = \iint_D \left[-e^x\sin(y) - (-e^x\sin(y))\right]\mathrm{d}A = 0.$$

Questions

1. Find the work done by $\mathbf{F} = x^2\mathbf{i} - 2yz\mathbf{j} + z\mathbf{k}$ in moving an object along the line segment from (1, 1, 1) to (4, 4, 4).
2. A particle moves once counterclockwise around the circle of radius 6 about the origin, under the influence of the force $\mathbf{F} = (e^x - y + x\cosh(x))\mathbf{i} + (y^{3/2} + x)\mathbf{j}$ Calculate the work done.
3. If $\mathbf{F} = 4y\mathbf{i} + 6x\mathbf{j}$ and C is given by $x^2 + y^2 = 1$, evaluate $\oint_C \mathbf{F}.\mathrm{d}\mathbf{r}$ in two different ways, that is direct method and the Green's theorem.

Line Integral With Respect to Arc Length

- In some contexts it is useful to have a line integral with respect to arc length along C. If $\phi(x, y, z)$ is a **scalar field** and C is a smooth curve with coordinate functions $x = x(t), y = y(t), z = z(t)$ for $a \leq t \leq b$, we define

$$\int_C \phi(x, y, z) \mathrm{d}s = \int_a^b \phi\left(x(t), y(t), z(t)\right) \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2}$$

- Example
- Evaluate the line integral $\int_c (x + y) \mathrm{d}s$ with C given by $x = y = t, z = t^2$ for $0 \leq t \leq 2$.
  Solution

$$\begin{aligned} x &= t, \quad dx = dt, \\ y &= t, \quad dy = dt, \\ z &= t^2, \quad dz = 2tdt. \end{aligned}$$

[ ]

Mass and Centre of Mass the Wire

- Suppose the density of wire is $\delta(x, y, z)$ at point (x, y, z) hence

  mass of the wire, $m = \int_c \delta(x, y, z)\mathrm{d}s,$ where $ds = \sqrt{[x'(t)]^2 + [y'($

- The coordinates $(\tilde{x}, \tilde{y}, \tilde{z})$ of the centre of mass of the wire is given as

- $$\tilde{x} = \frac{1}{m}\int_c x\delta(x, y, z)\mathrm{d}s,$$
  $$\tilde{y} = \frac{1}{m}\int_c y\delta(x, y, z)\mathrm{d}s,$$
  $$\tilde{z} = \frac{1}{m}\int_c z\delta(x, y, z)\mathrm{d}s.$$

Example

- A wire is bent into the shape of the quarter circle C given by $x = 2\cos(t), y = 2\sin(t), z = 3$ for $0 \leq t \leq \pi/2$. The density function is $\delta(x, y, z) = xy^2$. Find the mass and centre of mass of the wire.

Evaluation of Double Integrals

- **Theorem: Fubini's Theorem**
  Let f be continuous on a region R.
- (i) If R is of Type I, $R: a \le x \le b,\ g_1(x) \le y \le g_2(x)$, then

$$\iint\limits_R f(x,y)\mathrm{d}A = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\mathrm{d}y\mathrm{d}x.$$

- (ii) If R is of Type II, $R: c \le y \le d,\ h_1(y) \le x \le h_2(y)$, then

$$\iint\limits_R f(x,y)\mathrm{d}A = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)\mathrm{d}x\mathrm{d}y.$$

Examples

- Evaluate the double integral over the region R that is bounded by the graphs of the given equations. Choose the most convenient order of integration.
  1. $\iint\limits_R (2x+4y+1)\mathrm{d}A;\ y = x^2, y = x^3.$
- **Solution**
- Sketch the region; Using Type I

$$\iint\limits_R (2x+4y+1)\mathrm{d}A = \int_0^1 \int_{y=x^3}^{y=x^2} (2x+4y+1)\mathrm{d}y\mathrm{d}x = \int_0^1 \left[\int_{y=x^3}^{y=x^2} (2x\right.$$

$$= \int_0^1 \left[2xy+2y^2+y\right]\Big|_{x^3}^{x^2},$$

$$= \int_0^1 \left(x^3+x^2-2x^6\right)\mathrm{d}x,$$

$$= \frac{x^4}{4}+\frac{x^3}{3}-\frac{2x^7}{7}\Big|_0^1,$$

- 2. $\iint\limits_R x^3y^2\mathrm{d}A$; $y=x, y=0, x=1$.
- **Solution**
- Sketch the region; Using Type II

$$\iint\limits_R x^3y^2\mathrm{d}A = \int_0^1 \int_{x=y}^{x=1} x^3y^2\mathrm{d}x\mathrm{d}y = \int_0^1 \left[\int_{x=y}^{x=1} x^3\mathrm{d}x\right] y^2\mathrm{d}y,$$

$$= \int_0^1 \left(\frac{x^4}{4}\right)\Big|_{x=y}^{x=1} y^2\mathrm{d}y,$$

$$= \int_0^1 \left[\frac{1}{4}-\frac{y^4}{4}\right] y^2\mathrm{d}y = \frac{1}{4}\int_0^1 \left(y^2-y^6\right)\mathrm{d}y,$$

$$= \frac{1}{4}\left(\frac{y^3}{3}-\frac{y^7}{7}\right)\Big|_0^1,$$

$$= \left(\frac{1}{3}-\frac{1}{7}\right)-0,$$

$$\iint x^3y^2\mathrm{d}A \quad 1$$

Evaluation of Triple Integrals

- **Evaluation by Iterated Integrals** If the region D is bounded above by the graph of $z = f_2(x,y)$ and bounded below by the graph of $z = f_1(x,y)$, and if R is a Type I region, then, the triple integral of F over D can be written as an iterated integral:
- $$\iiint\limits_D F(x,y,z)\mathrm{d}V = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{f_1(x,y)}^{f_2(x,y)} F(x,y,z)\mathrm{d}z\mathrm{d}y\mathrm{d}x.$$

Example

- Evaluate the iterated integral $\int_0^6 \int_0^{6-x} \int_0^{6-x-y} \mathrm{d}z\mathrm{d}y\mathrm{d}x$.
- **Solution**
- There are six possible orders of integration: $\mathrm{d}z\mathrm{d}y\mathrm{d}x, \mathrm{d}x\mathrm{d}y\mathrm{d}z, \mathrm{d}z\mathrm{d}x\mathrm{d}y, \mathrm{d}x\mathrm{d}z\mathrm{d}y, \mathrm{d}y\mathrm{d}x\mathrm{d}z, \mathrm{d}y\mathrm{d}z\mathrm{d}x$.

$$
\begin{aligned}
\int_0^6 \int_0^{6-x} \int_0^{6-x-y} \mathrm{d}z\mathrm{d}y\mathrm{d}x &= \int_0^6 \int_0^{6-x} \left(\int_0^{6-x-y} \mathrm{d}z \right) \mathrm{d}y\mathrm{d}x, \\
&= \int_0^6 \int_0^{6-x} \left(z\Big|_0^{6-x-y} \right) \mathrm{d}y\mathrm{d}x, \\
&= \int_0^6 \int_0^{6-x} (6-x-y)\mathrm{d}y\mathrm{d}x = \int_0^6 \left(\int_0^{6-x} (6 \right. \\
&= \int_0^6 \left(6y - xy - \frac{y^2}{2} \right) \Bigg|_0^{6-x} \mathrm{d}x, \\
&= \int_0^6 \left[6(6-x) - x(6-x) - \frac{(6-x)^2}{2} \right] \mathrm{d}x =
\end{aligned}
$$

Green's Theorem

- Theorem: Green's Theorem in the Plane
- Suppose that C is a piecewise-smooth simple closed curve bounding a simply connected region R. If P, Q, $\partial P/\partial y$, and $\partial Q/\partial x$ are continuous on R, then
- $$\oint_C P\mathrm{d}x + Q\mathrm{d}y = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathrm{d}A.$$
- Example
  Use Green's Theorem to evaluate the line integral $\oint_C \left(x^5 + 3y\right)\mathrm{d}x + \left(2x - e^{y^3}\right)\mathrm{d}y$, where C is the circle $(x-1)^2 + (y-5)^2 = 4$.

Solution

- $P = x^5 + 3y$ and $Q = 2x - e^{y^3}$ Thus we have, $\frac{\partial P}{\partial y} = 3$ and $\frac{\partial Q}{\partial x} = 2$
  $\oint_C \left(x^5 + 3y\right) \mathrm{d}x + \left(2x - e^{y^3}\right) \mathrm{d}y = \iint\limits_R (2-3)\mathrm{d}A = -\iint\limits_R \mathrm{d}A,$
- Now the double integral $\iint\limits_R dA$ gives the area of the region R bounded by the circle of radius 2. Since the area of the circle is $\pi r^2 = \pi . 2^2 = 4\pi$,

$$\therefore \oint_C \left(x^5 + 3y\right) \mathrm{d}x + \left(2x - e^{y^3}\right) \mathrm{d}y = -4\pi.$$

Questions

1. Use Green's theorem to evaluate the line integral, $\oint_C 2y\,\mathrm{d}x + 5x\,\mathrm{d}y$ where C is the circle $(x-1)^2 + (y+3)^2 = 25$.
2. Evaluate $\oint_C \mathbf{F}.\mathrm{d}\mathbf{r}$ counterclockwise around the boundary C of the region R by Green's theorem, where $\mathbf{F} = [y, -x]$, C the circle $x^2 + y^2 = \frac{1}{4}$.

Periodic Functions

- $f(\theta) = \sin(\theta + 2\pi n) = \sin\theta, \quad f(\theta) = \cos(\theta + 2\pi n) = \cos\theta$ where n is any positive integer.
  Functions with this kind of repetitive behaviour are called periodic functions.
- Definition 1
  A function f is called **periodic** if there is a positive number p such that, whenever θ is in the domain of f, so is $\theta + p$, and $f(\theta + p) = f(\theta)$.
  If there is a smallest such number p, this smallest value is called the **fundamental period** of f.
- Definition 2
  A function f is **periodic** if there exists a positive real number p such that $f(x + p) = f(x)$ for all x in the domain of f. The smallest such positive p, if it exists, is called the **fundamental period of** f (or often just the **period of** f).

Trigonometric series

- Trigonometric series Is a series in cosines and sines of multiple angles, that is, a series of the form

$$\frac{a_0}{2}+\sum_{n=1}^{\infty}(a_n\cos nx+b_n\sin nx)$$

  where a_n, and b_n are called the coefficients of the trigonometric series.

- **Even and odd functions**
  If f $(-\theta)=f(\theta)$ for all θ in the domain of f, then f is an **even function.**
  If $f(-\theta)=-f(\theta)$ for all θ in the domain of f, then f is an **odd function.**
  OR
  If f $(-x)=f(x)$ for all x in the domain of f, then f is an **even function.**

- If $f(-x) = -f(x)$ for all x in the domain of f, then f is an **odd function.**
- The trigonometric cosine function is an even function while the sine function is an odd function.
  therefore $\cos(-\theta) = \cos\theta$ and $\sin(-\theta) = -\sin\theta$

Example

- Determine whether each of the following function is even, odd, or neither even nor odd.
  (a) $f(x) = x$
  (b) $f(x) = x^2$
  (c) $f(x) = x^3$
  (d) $f(x) = \pi - x$
  Solution
  (a) Given that

$$f(x) = x,$$

$$f(-x) = -x, \quad \text{substitute} -x \quad \text{for} \quad f(x)$$

Solution cont...

- (b) Given that

$$
\begin{aligned}
f(x) &= x^2, \\
f(-x) &= (-x)^2 = (-x)(-x), \quad \text{substitute} \quad -x \quad \text{for} \quad f(x) \\
&= x^2, \\
f(-x) &= f(x)
\end{aligned}
$$

Since $f(-x) = f(x)$, f is an even function.

Solution cont...

- (c) Given that

$$
\begin{aligned}
f(x) &= x^3, \\
f(-x) &= (-x)^3 = (-x)(-x)(-x), \quad \text{substitute} \quad -x \quad \text{for} \quad f(x) \\
&= -x^3, \\
f(-x) &= -f(x)
\end{aligned}
$$

Since $f(-x) = -f(x)$, f is an odd function.
(d) Given that

$$
\begin{aligned}
f(x) &= \pi - x, \\
f(-x) &= \pi - (-x) = \pi + x, \quad \text{substitute} - x \quad \text{for} \quad f(x) \\
-f(x) &= -(\pi - x) = -\pi + x.
\end{aligned}
$$

Since $f(-x) \neq f(x)$ and $f(-x) \neq -f(x)$, Hence f is neither even nor odd

Properties of Even/Odd Functions

(a). The product of two even functions is even.

(b) . The product of two odd functions is even.

(c). The product of an even function and an odd function is odd.

(d). The sum (difference) of two even functions is even.

(e). The sum (difference) of two odd functions is odd.

(f). If f is even $[f(-x) = f(x)]$, then $\int_{-a}^{a} f(x)\mathrm{d}x = 2\int_{0}^{a} f(x)\mathrm{d}x$.

(g). If f is odd $[f(-x) = -f(x)]$, then $\int_{-a}^{a} f(x)\mathrm{d}x = 0$.

Examples

1. $\int_{-5}^{5} x^2\mathrm{d}x = 2\int_{0}^{5} x^2\mathrm{d}x = 2.\frac{x^3}{3}\Big|_{0}^{5} = \frac{250}{3}$.
2. $\int_{-2}^{2} x\mathrm{d}x = \int_{-2}^{2} x\mathrm{d}x = \frac{x^2}{2}\Big|_{-2}^{2} = \frac{2^2}{2} - \frac{(-2)^2}{2} = \frac{4}{2} - \frac{4}{2} = 0$.

Definitions

- **Fourier Series** is an infinite series of trigonometric functions used to represent a given periodic function.
- **Fourier Series Function**
  The **Fourier series** of a function f defined on the interval $[-L, L]$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{\pi n x}{L}\right) + b_n \sin\left(\frac{\pi n x}{L}\right) \right],$$

  Where a_n and b_n are Fourier coefficients defined as

$$a_0 = \frac{1}{L}\int_{-L}^{L} f(x)\mathrm{d}x$$
$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos(\frac{\pi n x}{L})\mathrm{d}x \quad \text{for } n = 1,2,3,....$$
$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin(\frac{\pi n x}{L})\mathrm{d}x \quad \text{for } n = 1,2,3,....$$

Example 1.

- Compute the Fourier series of $f(x) = x$ on $[-1,1]$.

  Solution

  Since $f(x) = x$ is odd, then $a_0 = 0$ and $a_n = 0$.

$$b_n = \frac{1}{1}\int_{-1}^{1} x\sin\left(\frac{\pi nx}{1}\right)\mathrm{d}x = 2\int_{0}^{1} x\sin(\pi nx)\mathrm{d}x,$$

  By Using Integration by Parts,

$$b_n = \frac{-2x}{\pi n}\cos(\pi nx)\Big|_0^1 = \frac{-2}{\pi n}\cos(\pi n) - 0 = \frac{-2(-1)^n}{\pi n} = \frac{2(-1)^1(-1)^n}{\pi n} =$$

  Therefore, the Fourier series is

$$f(x) = \sum_{n=1}^{\infty} \frac{2((-1)^{n+1})}{\pi n}\sin(\pi nx).$$

Example 2.

- Expand $f(x) = \begin{cases} 0 & \text{if } -\pi < x < 0 \\ \pi - x & \text{if } 0 \le x < \pi \end{cases}$ in a Fourier series.
- Solution

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\mathrm{d}x = \frac{1}{\pi}\left[\int_{-\pi}^{0} 0\mathrm{d}x + \int_{0}^{\pi}(\pi - x)\mathrm{d}x\right] = 0 + \frac{1}{\pi}(\pi x - \frac{x^2}{2})\Big|_0^{\pi}$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(\frac{\pi n x}{\pi})\mathrm{d}x = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\mathrm{d}x = \frac{1}{\pi}\left[\int_{-\pi}^{0} 0\mathrm{d}x + \int_{0}\right.$$

$$a_n = \frac{1}{\pi}\int_{0}^{\pi}(\pi - x)\cos(nx)\mathrm{d}x,$$

- By using Integration by Parts, ILATE, differentiate the first to appear

$$\int u\mathrm{d}v = uv - \int v\mathrm{d}u$$

$$u = \pi - x \quad \mathrm{d}u = -\mathrm{d}x$$

$$dv = \cos nx\mathrm{d}x \quad v = \frac{1}{n}\sin nx, \quad \text{thus}$$

$$a_n = \frac{1}{\pi}\left[(\pi - x)\frac{\sin nx}{n}\Big|_0^{\pi} + \frac{1}{n}\int_0^{\pi}\sin nx\mathrm{d}x\right] = \frac{1}{\pi}\left[0 + \frac{1}{n}\int_0^{\pi}\sin nx\mathrm{d}x\right]$$

$$= \frac{1}{\pi n}\int_0^{\pi}\sin nx\mathrm{d}x,$$

$$= -\frac{1}{\pi n}\frac{\cos nx}{n}\Big|_0^{\pi} = -\frac{1}{\pi n^2}\cos nx\Big|_0^{\pi} = -\frac{1}{\pi n^2}(\cos n\pi - 1) = \frac{-\cos}{\pi}$$

$$a_n = \frac{1 - \cos n\pi}{\pi n^2} \text{ but } \cos(\pi n) = (-1)^n,$$

$$\therefore a_n = \frac{1 - (-1)^n}{\pi n^2}.$$

- In Similar manner

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(\frac{\pi n x}{\pi})\mathrm{d}x = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\mathrm{d}x,$$

$$b_n = \frac{1}{\pi}\left[\int_{-\pi}^{0} 0\sin(nx)\mathrm{d}x + \int_{0}^{\pi}(\pi - x)\sin(nx)\mathrm{d}x\right] = \frac{1}{\pi}\int_{0}^{\pi}(\pi - x)\sin(nx)$$

- By using Integration by Parts

$$\int u \mathrm{d}v = uv - \int v \mathrm{d}u$$

$$u = \pi - x \quad \mathrm{d}u = -\mathrm{d}x$$

$$dv = \sin nx \mathrm{d}x \quad v = -\frac{1}{n}\cos nx, \quad \text{thus}$$

$$b_n = \frac{1}{\pi}\left[-(\pi x)\frac{\cos nx}{n}\bigg|_0^{\pi} + \frac{1}{n}\int_0^{\pi}\cos nx \mathrm{d}x\right],$$

$$= \frac{1}{\pi}\left[0 + \frac{\pi}{n} + \frac{1}{n^2}\sin nx\bigg|_0^{\pi}\right],$$

$$= \frac{1}{\pi}\left(\frac{\pi}{n} + 0\right),$$

$$\therefore b_n = \frac{1}{n},$$

- Therefore, the Fourier series is

$$f(x) = \frac{\pi}{4} + \sum_{n=1}^{\infty} \left[\frac{1-(-1)^n}{\pi n^2} \cos(nx) + \frac{1}{n} \sin(nx) \right].$$

Questions

1. Determine whether the given function is even, odd, or neither.

$$\text{(i) } f(x) = x - x^2 \quad \text{(ii) } f(x) = e^x \quad \text{(iii) } f(x) = x\cos x.$$

2. Find the Fourier series of the function f on the given interval.

$$f(x) = \begin{cases} 0, & -\pi < x < 0 \\ 1, & 0 \le x < \pi \end{cases}$$

3. Compute the Fourier series of $f(x) = x$ on $[-\pi, \pi]$.
4. Compute the Fourier series of $f(x) = e^x$ on $[-1, 1]$.

Even and odd Fourier series representation

- Definition
  Fourier Cosine and Sine Series
- (i) The Fourier series of an even function on the interval $(-L, L)$ is the **cosine series**

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{\pi n x}{L}\right), \text{ where}$$

$$a_0 = \frac{2}{L}\int_0^L f(x)\mathrm{d}x,$$

$$a_n = \frac{2}{L}\int_0^L f(x)\cos(\frac{\pi n x}{L})\mathrm{d}x.$$

- (ii) The Fourier series of an odd function on the interval $(-L, L)$ is the **sine series**

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{\pi n x}{L}\right), \text{ where}$$

$$b_n = \frac{2}{L}\int_0^L f(x)\sin(\frac{\pi n x}{L})\mathrm{d}x.$$

Example

- Find the Fourier sine series of $f(x) = x, -2 < x < 2$.
- **Solution**
  $b_n = \frac{2}{2}\int_0^2 x\sin(\frac{\pi n x}{2})\mathrm{d}x = \int_0^2 x\sin(\frac{\pi n x}{2})\mathrm{d}x,$

Solution cont...

- By using Integration by Parts,

$$\int u\mathrm{d}v = uv - \int v\mathrm{d}u$$

$$u = x \quad \mathrm{d}u = \mathrm{d}x$$

$$dv = \sin\frac{\pi nx}{2}\mathrm{d}x \quad v = -\frac{2}{\pi n}\cos\frac{\pi nx}{2}, \text{ thus}$$

$$b_n = \left.\frac{-2x}{\pi n}\cos\frac{\pi nx}{2}\right|_0^2 + \int_0^2 \frac{-2}{\pi n}\cos\frac{\pi nx}{2}\mathrm{d}x,$$

$$= \frac{-4}{\pi n}\cos(\pi n) + 0 + \left.\frac{-4}{\pi^2 n^2}\sin(\frac{\pi nx}{2})\right|_0^2 = \frac{-4}{\pi n}(-1)^n + 0 = \frac{4(-1)(}{\pi r}$$

$$b_n = \sum_{n=1}^{\infty} \frac{4(-1)^{n+1}}{\pi n}\sin(\frac{\pi nx}{2}),$$

- Therefore, the sine series is $f(x) = \frac{4}{\pi}\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}\sin(\frac{\pi n x}{2})$.

Questions

1. Write the Fourier cosine and sine series for f on the interval: $f(x) = 4, 0 \leq x \leq 3$.
2. Expand the given function in an appropriate cosine or sine series: $f(x) = x^2, -1 < x < 1$.

Complex exponential form of Fourier series

Definition

The **complex Fourier series** of functions *f* defined on an interval $(-L, L)$ is given by

$$f(x) = \sum_{n=-\infty}^{\infty} d_n e^{i\pi n x/L}, \text{ where}$$

$$d_n = \frac{1}{2L}\int_{-L}^{L} f(x) e^{-i\pi n x/L}\mathrm{d}x, n = 0, \pm 1, \pm 2, \ldots.$$

Example

- Expand $f(x) = e^{-x}, -\pi < x < \pi$, in a complex Fourier series.

Solution

$$d_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-x} e^{-i\pi nx/}\pi \mathrm{d}x = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-x} e^{-inx}\mathrm{d}x = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-(1+in)x}\mathrm{d}x$$

$$= -\frac{1}{2\pi(1+in)} e^{-(1+in)x}\Big|_{-\pi}^{\pi},$$

$$= -\frac{1}{2\pi(1+in)}\left[e^{-(1+in)\pi} - \left(e^{(1+in)\pi}\right)\right],$$

$$d_n = \frac{1}{2\pi(1+in)}\left[e^{(1+in)\pi} - \left(e^{-(1+in)\pi}\right)\right],$$

Using Euler's Formula $e^{ix} = \cos x + i\sin x, \quad e^{-ix} = \cos x - i\sin x$ to simplify the coefficients d_n,

$$e^{(1+in)\pi} = e^{\pi}\left[\cos(\pi n) + i\sin(\pi n)\right] = (-1)^n e^{\pi},$$

$$e^{-(1+in)\pi} = e^{-\pi}\left[\cos(\pi n) - i\sin(\pi n)\right] = (-1)^n e^{-\pi}.$$

Question

1. Let $f(x) = x, \text{for} -1 \leq x < 1$. Find the complex Fourier series of f.
2. Find the complex Fourier series of f on the given interval:

$$f(x) = \begin{cases} 0, & \frac{-1}{2} < x < 0 \\ 1, & 0 < x < \frac{1}{4} \\ 0, & \frac{1}{4} < x < \frac{1}{2} \end{cases}$$

Fourier Integrals and Transforms

- **Fourier Integrals**

Definition

The **Fourier integral** of a function f defined on the interval $(-\infty, \infty)$ is given by

$$f(x) = \frac{1}{\pi}\int_0^\infty [A(\alpha)\cos\alpha x + B(\alpha)\sin\alpha x]\mathrm{d}\alpha, \text{ where}$$
$$A(\alpha) = \int_{-\infty}^\infty f(x)\cos\alpha x\mathrm{d}x$$
$$B(\alpha) = \int_{-\infty}^\infty f(x)\sin\alpha x\mathrm{d}x.$$

Example

Find the Fourier integral representation of the piecewise-continuous function $f(x) = \begin{cases} 0, & x < 0 \\ 1, & 0 < x < 2 \\ 0, & x > 2 \end{cases}$

Fourier Integrals and Transforms cont...

- **Solution**

$$A(\alpha) = \int_{-\infty}^{\infty} f(x)\cos\alpha x\,\mathrm{d}x,$$
$$= \int_{-\infty}^{0} f(x)\cos\alpha x\,\mathrm{d}x + \int_{0}^{2} f(x)\cos\alpha x\,\mathrm{d}x + \int_{2}^{\infty} f(x)\cos\alpha x\,\mathrm{d}x = \int_{0}^{2}$$
$$A(\alpha) = \frac{\sin 2\alpha}{\alpha}.$$
$$B(\alpha) = \int_{-\infty}^{\infty} f(x)\sin\alpha x\,\mathrm{d}x,$$
$$= \int_{-\infty}^{0} f(x)\sin\alpha x\,\mathrm{d}x + \int_{0}^{2} f(x)\sin\alpha x\,\mathrm{d}x + \int_{2}^{\infty} f(x)\sin\alpha x\,\mathrm{d}x = \int_{0}^{2} \mathrm{si}$$
$$B(\alpha) = \frac{1-\cos 2\alpha}{\alpha}.$$

Therefore, the Fourier integral representation is
$f(x) = \frac{1}{\pi}\int_0^\infty [\left(\frac{\sin 2\alpha}{\alpha}\right)\cos\alpha x + \left(\frac{1-\cos 2\alpha}{\alpha}\right)\sin\alpha x]\mathrm{d}\alpha.$

Fourier Cosine Integrals, Sine Integrals, and Complex form

- Definitions: (i) The Fourier integral of an even function on the interval $(-\infty, \infty)$ is the **cosine integral**

$$f(x) = \frac{2}{\pi}\int_0^\infty A(\alpha)\cos\alpha x \,\mathrm{d}\alpha, \text{ where}$$
$$A(\alpha) = \int_0^\infty f(x)\cos\alpha x \,\mathrm{d}x$$

- (ii) The Fourier integral of an odd function on the interval $(-\infty, \infty)$ is the **sine integral**

$$f(x) = \frac{2}{\pi}\int_0^\infty B(\alpha)\sin\alpha x \,\mathrm{d}\alpha, \text{ where}$$
$$B(\alpha) = \int_0^\infty f(x)\sin\alpha x \,\mathrm{d}x$$

- (iii)The **Complex Fourier integral representation** of a function f is defined by

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} C(\alpha)e^{-i\alpha x}\mathrm{d}\alpha, \text{ where}$$
$$C(\alpha) = \int_{-\infty}^{\infty} f(x)e^{i\alpha x}\mathrm{d}x.$$

Examples

(1). Find the complex form of the Fourier integral for the function of

$$f(x) = \begin{cases} e^x, & |x| < 1 \\ 0, & |x| > 1. \end{cases}$$

Solution

$$C(\alpha) = \int_{-1}^{1} e^x e^{i\alpha x}\mathrm{d}x = \int_{-1}^{1} e^{(1+i\alpha)x}\mathrm{d}x = \frac{1}{1+i\alpha}e^{(1+i\alpha)x}\Big|_{-1}^{1} = \frac{e^{(1+}}{}$$

$\therefore$The complex form of the Fourier integral is

(2). Find the Fourier integral representation of the function
$f(x) = \begin{cases} 1, & |x| < a \\ 0, & |x| > a. \end{cases}$

Solution

Since f is an even function, hence we represent f by the Fourier cosine integral

$$\begin{aligned} A(\alpha) = \int_0^\infty f(x)\cos\alpha x \mathrm{d}x &= \int_0^a f(x)\cos\alpha x \mathrm{d}x + \int_a^\infty f(x)\cos\alpha x \mathrm{d}x, \\ &= \int_0^a \cos\alpha x \mathrm{d}x, \\ &= \frac{\sin a\alpha}{\alpha}. \end{aligned}$$

Therefore, The Fourier integral is $f(x) = \frac{2}{\pi}\int_0^\infty \frac{\sin a\alpha \cos\alpha x}{\alpha}\mathrm{d}\alpha.$

Fourier Transforms

- **Fourier Transforms** is a function derived from a given non-periodic function and representing it as a series of sinusoidal functions.

  Fourier Transform Pairs

- (i) Fourier Transform: $\mathscr{F}\{f(x)\} = \int_{-\infty}^{\infty} f(x)e^{i\alpha x}\mathrm{d}x = F(\alpha)$
  Inverse Fourier Transform:
  $\mathscr{F}^{-1}\{F(\alpha)\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\alpha)e^{-i\alpha x}\mathrm{d}\alpha = f(x)$
- (ii) Fourier Sine Transform: $\mathscr{F}_s\{f(x)\} = \int_0^{\infty} f(x)\sin\alpha x\mathrm{d}x = F(\alpha)$
  Inverse Fourier Sine Transform:
  $\mathscr{F}^{-1}{}_s\{F(\alpha)\} = \frac{2}{\pi}\int_0^{\infty} F(\alpha)\sin\alpha x\mathrm{d}\alpha = f(x)$
- (iii) Fourier Cosine Transform: $\mathscr{F}_c\{f(x)\} = \int_0^{\infty} f(x)\cos\alpha x\mathrm{d}x = F(\alpha)$
  Inverse Fourier Cosine Transform:
  $\mathscr{F}^{-1}{}_c\{F(\alpha)\} = \frac{2}{\pi}\int_0^{\infty} F(\alpha)\cos\alpha x\mathrm{d}\alpha = f(x)$.

Fourier Transforms of Derivatives

- **Fourier Transform**
  If $f(x) \to 0$ as $x \to \pm\infty$, then integration by parts gives
  $\mathscr{F}\{f'(x)\} = -i\alpha F(\alpha)$.
  Also
  $\mathscr{F}_s\{f''(x)\} = -\alpha^2 F(\alpha)$.
  Generally, the Fourier transform of the nth derivative is

$$\mathscr{F}\{f^{(n)}(x)\} = (-i\alpha)^n \mathscr{F}\{f(x)\} = (-i\alpha)^n F(\alpha) \quad \text{for} \quad n = 0, 1, 2, ...$$

- **Fourier Sine Transform**

$$\mathscr{F}_s\{f''(x)\} = -\alpha^2 F(\alpha) + \alpha f(0).$$

- **Fourier Cosine Transform**

$$\mathscr{F}_c\{f''(x)\} = -\alpha^2 F(\alpha) - f'(0).$$

N.B:

- **How do we know which transform to use on a given boundary-value problem?**
  To use the **Fourier transform** the domain of the variable to eliminate must be $(-\infty, \infty)$.
  To utilize a **sine** or **cosine transform**, the domain of at least one of the spatial variables in the problem must be $[0, \infty)$. However, the determining factor in choosing between the sine transform and the cosine transform is the type of boundary condition specified at $x = 0$ (or $y = 0$), that is, whether u or its first partial derivative is given at this boundary.

- If we let u denote the dependent variable and x and y the independent variables, then the general form of a **linear second-order partial differential equation** is given by

$$A\frac{\partial^2 u}{\partial x^2}+B\frac{\partial^2 u}{\partial x\partial y}+C\frac{\partial^2 u}{\partial y^2}+D\frac{\partial u}{\partial x}+E\frac{\partial u}{\partial y}+Fu=G,$$

where the coefficients $A, B, C, ..., G$ are constants or functions of x and y. When $G(x,y)=0$, is said to be **homogeneous**; otherwise, it is **non-homogeneous.**

Linear Partial Differential Equation

- **Classical PDEs and Boundary-Value Problems(BVPs)**

  The **one-dimensional heat equation**: $k\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, k > 0$

  The **one- dimensional wave equation**: $a^2\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}$

  The **Laplace's equation in two dimensions**: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

  "One-dimensional" refers to the fact that x denotes a spatial(space) dimension whereas t represents time, "two dimensional" means that x and y are both spatial dimensions.

  Using Fourier Series to Solve PDE

Examples

- 1. Finding the temperature $u(x,t)$ in an infinite rod. Solve the heat equation $k\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, -\infty < x < \infty, t > 0$, subject to $u(x,0) = f(x)$ where $f(x) = \begin{cases} u_0, & |x| < 1 \\ 0, & |x| > 1. \end{cases}$
- **Solution**
  Step 1. To find the Fourier transform of the PDE.
  Because the domain of x is the infinite interval $(-\infty,\infty)$ then we use the Fourier transform and define the transform of $u(x,t)$ to be

$$\mathscr{F}\{u(x,t)\} = \int_{-\infty}^{\infty} u(x,t)e^{i\alpha x}\mathrm{d}x = U(\alpha,t)$$

If we write

- $$\mathscr{F}\left\{\frac{\partial^2 u}{\partial x^2}\right\} = -\alpha^2 U(\alpha, t) \text{ and } \mathscr{F}\left\{\frac{\partial u}{\partial t}\right\} = \frac{d}{dt}\mathscr{F}\{u(x,t)\} = \frac{dU}{dt}$$

  Thus $\mathscr{F}\left\{k\frac{\partial^2 u}{\partial x^2}\right\} = \mathscr{F}\{\frac{\partial u}{\partial t}\} \Rightarrow -k\alpha^2 U(\alpha, t) =$
  $\frac{dU}{dt}$ or $\frac{dU}{dt} + k\alpha^2 U(\alpha, t) = 0$
- **Step 2. To solve the ODE formed after transforming the PDE.** Solving $\frac{dU}{dt} + k\alpha^2 U(\alpha, t) = 0$ gives $U(\alpha, t) = ce^{-k\alpha^2 t}$.
- **Step 3. To find the Fourier Transform of the initial condition and write the solution.** The initial temperature $u(x,0) = f(x) = c$ in the rod and its Fourier transform is

$$\mathscr{F}\{u(x,0)\} = U(\alpha,0) = \int_{-\infty}^{\infty} f(x)e^{i\alpha x}\mathrm{d}x = \int_{-1}^{1} u_0 e^{i\alpha x}\mathrm{d}x = u_0\frac{(e^{i\alpha} - e^{-}}{i\alpha}$$

- By Using Euler's formula to simplify gives

$$U(\alpha,0) = 2u_0 \frac{\sin\alpha}{\alpha} \quad \text{then} \quad U(\alpha,t) = 2u_0 \frac{\sin\alpha}{\alpha} e^{-k\alpha^2 t}.$$

- **Step 4. To find the inverse Fourier transform of the solution.** Finding the inverse Fourier transform gives

$$u(x,t) = \frac{u_0}{\pi} \int_{-\infty}^{\infty} \frac{\sin\alpha}{\alpha} e^{-k\alpha^2 t} e^{-i\alpha x} \mathrm{d}\alpha.$$

Simplifying the integral by Euler's formula,

$$e^{-i\alpha x} = \cos \alpha x - i \sin \alpha x.$$

$$u(x,t) = \frac{u_0}{\pi}\int_{-\infty}^{\infty} e^{-k\alpha^2 t}\frac{\sin \alpha}{\alpha}\cos \alpha x \mathrm{d}\alpha - \frac{u_0}{\pi}\int_{-\infty}^{\infty} i e^{-k\alpha^2 t}\frac{\sin \alpha}{\alpha}\sin \alpha x \mathrm{d}\alpha,$$

$$u(x,t) = \frac{u_0}{\pi}\int_{-\infty}^{\infty} \frac{\sin \alpha \cos \alpha x}{\alpha} e^{-k\alpha^2 t}\mathrm{d}\alpha - 0,$$

$$\therefore u(x,t) = \frac{u_0}{\pi}\int_{-\infty}^{\infty} \frac{\sin \alpha \cos \alpha x}{\alpha} e^{-k\alpha^2 t}\mathrm{d}\alpha.$$

- 2. Solve the given boundary-value problem by an appropriate integral transform:

$$k\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, \quad -\infty < x < \infty, \quad t > 0 \quad u(x,0) = \begin{cases} 0, & x < 0 \\ e^{-x}, & x > 0. \end{cases}$$

Solution

Step 1. To find the Fourier transform of the PDE.

$$\mathscr{F}\left\{k\frac{\partial^2 u}{\partial x^2}\right\} = \mathscr{F}\{\frac{\partial u}{\partial t}\},$$

$$k\mathscr{F}\left\{\frac{\partial^2 u}{\partial x^2}\right\} = \mathscr{F}\{\frac{\partial u}{\partial t}\},$$

$$-k\alpha^2 U(\alpha,t) = \frac{dU(\alpha,t)}{dt} \quad \Rightarrow \frac{dU}{dt} = -\alpha^2 kU.$$

- **Step 2.** To solve the ODE: $\frac{dU}{dt} = -\alpha^2 kU$
  Using the method of separation of variables, we have

$$\frac{dU}{U} = -\alpha^2 kdt$$

Integrate both sides

$$\int \frac{dU}{U} = \int -\alpha^2 kdt = -\alpha^2 k \int dt,$$

$$\ln U = -\alpha^2 kt + A \quad \Rightarrow U = e^{-\alpha^2 kt + A} = e^A e^{-\alpha^2 kt} = Ce^{-\alpha^2 kt},$$

$$\therefore U(\alpha, t) = Ce^{-\alpha^2 kt}.$$

- **N.B:**
  When $t = 0$, $U(\alpha, 0) = C$, to get C we need to find Fourier transform of the initial condition.
- **Step 3.** To find the Fourier transform of the initial condition and write the solution.

$$
\begin{aligned}
\mathscr{F}\{u(x,0)\} &= \int_0^\infty e^{-x} e^{i\alpha x} \mathrm{d}x, \\
U(\alpha,0) &= \int_0^\infty e^{-(1-i\alpha)x} \mathrm{d}x, \\
&= \frac{1}{-(1-i\alpha)} e^{-(1-i\alpha)} \bigg|_0^\infty, \\
&= \frac{-1}{(1-i\alpha)}(0-1) = \frac{1}{1-i\alpha}, \\
U(\alpha,0) &= \frac{1+i\alpha}{1+\alpha^2}, \\
\therefore U(\alpha,t) &= \frac{1+i\alpha}{1+\alpha^2} e^{-\alpha^2 kt}.
\end{aligned}
$$

- **Step 4.** To find the inverse Fourier transform of the solution.

$$\mathscr{F}^{-1}\{U(\alpha,t)\} = \mathscr{F}^{-1}\left\{\frac{1+i\alpha}{1+\alpha^2}e^{-\alpha^2kt}\right\},$$

$$\begin{aligned}
u(x,t) &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{1+i\alpha}{1+\alpha^2}e^{-\alpha^2kt}e^{-i\alpha x}\mathrm{d}\alpha = \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{(1+i\alpha)e^{-i\alpha x}}{1+\alpha^2} \\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{(1+i\alpha)(\cos(\alpha x)-i\sin(\alpha x))}{1+\alpha^2}e^{-\alpha^2kt}\mathrm{d}\alpha, \\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{\cos(\alpha x)-i\sin(\alpha x)+i\alpha\cos(\alpha x)+\alpha\sin(\alpha x)}{1+\alpha^2}e \\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{\cos(\alpha x)-0+0+\alpha\sin(\alpha x)}{1+\alpha^2}e^{-\alpha^2kt}\mathrm{d}\alpha, \\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{\cos(\alpha x)+\alpha\sin(\alpha x)}{1+\alpha^2}e^{-\alpha^2kt}\mathrm{d}\alpha, \\
u(x,t) &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{\cos(\alpha x)+\alpha\sin(\alpha x)}{1+\alpha^2}e^{-\alpha^2kt}\mathrm{d}\alpha.
\end{aligned}$$

- $\therefore$ The solution of the heat equation is

$$u(x,t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{\cos(\alpha x) + \alpha \sin(\alpha x)}{1+\alpha^2} e^{-\alpha^2 kt} \mathrm{d}\alpha.$$

- Question
  1. Use an appropriate integral transform to solve:

$$k\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, \quad -\infty < x < \infty, \quad t > 0 \quad u(x,0) = \begin{cases} 0, & x < 0 \\ u_0, & 0 < x < \pi \\ 0, & x > \pi. \end{cases}$$